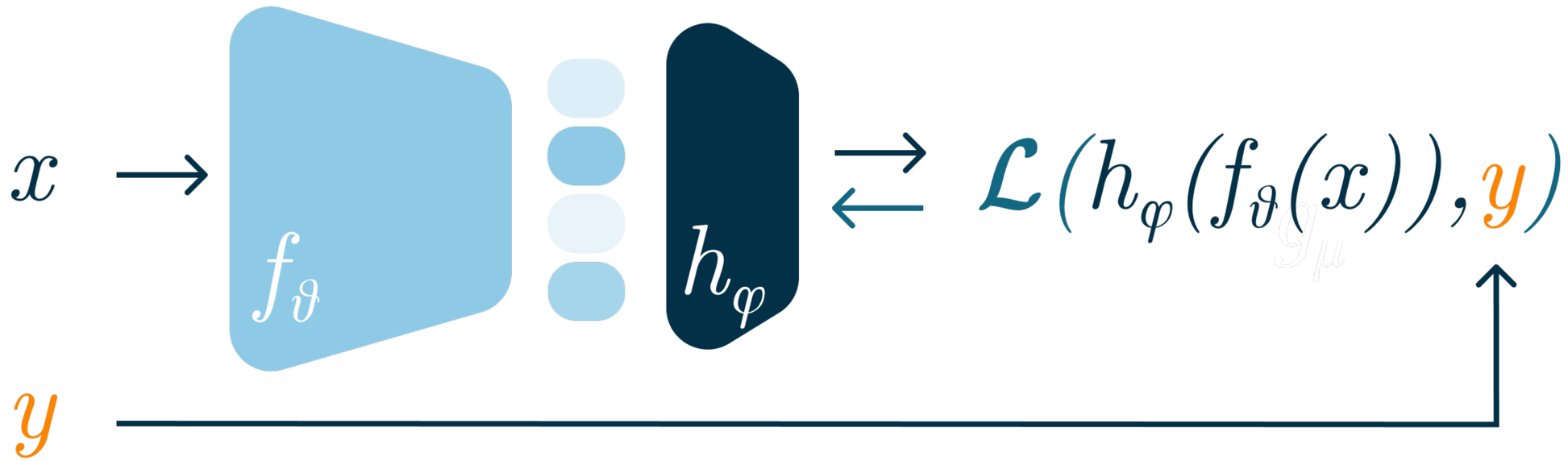
Capturing the Geometry of Self-Supervised Representations

A tale of minimum spanning trees and dimension estimation

PhD Defense
Palaiseau, September 21st, 2026

Julie Mordacq

Why self-supervised learning?



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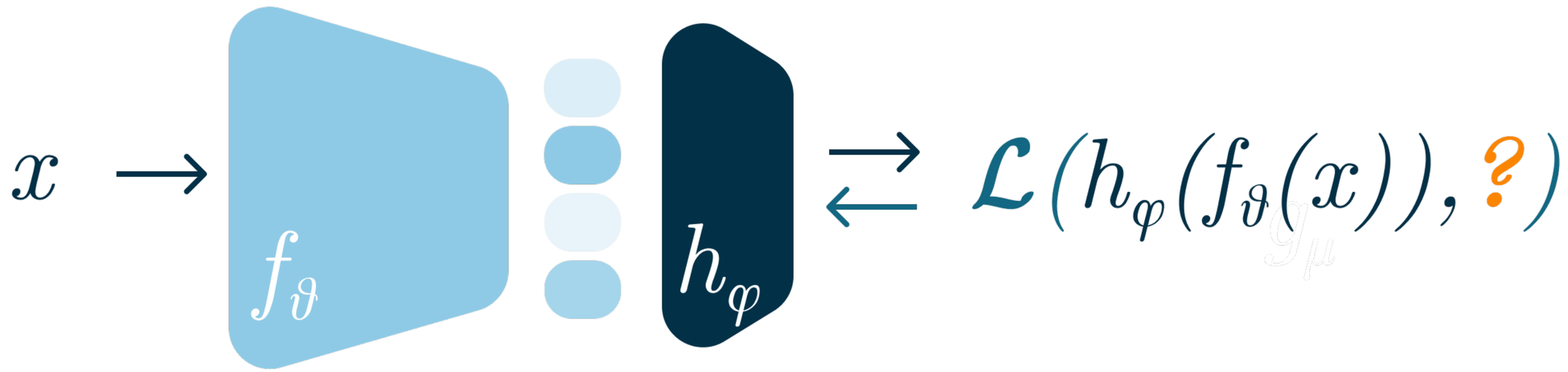

Why self-supervised learning?

Labels are limiting:
expensive & ambiguous

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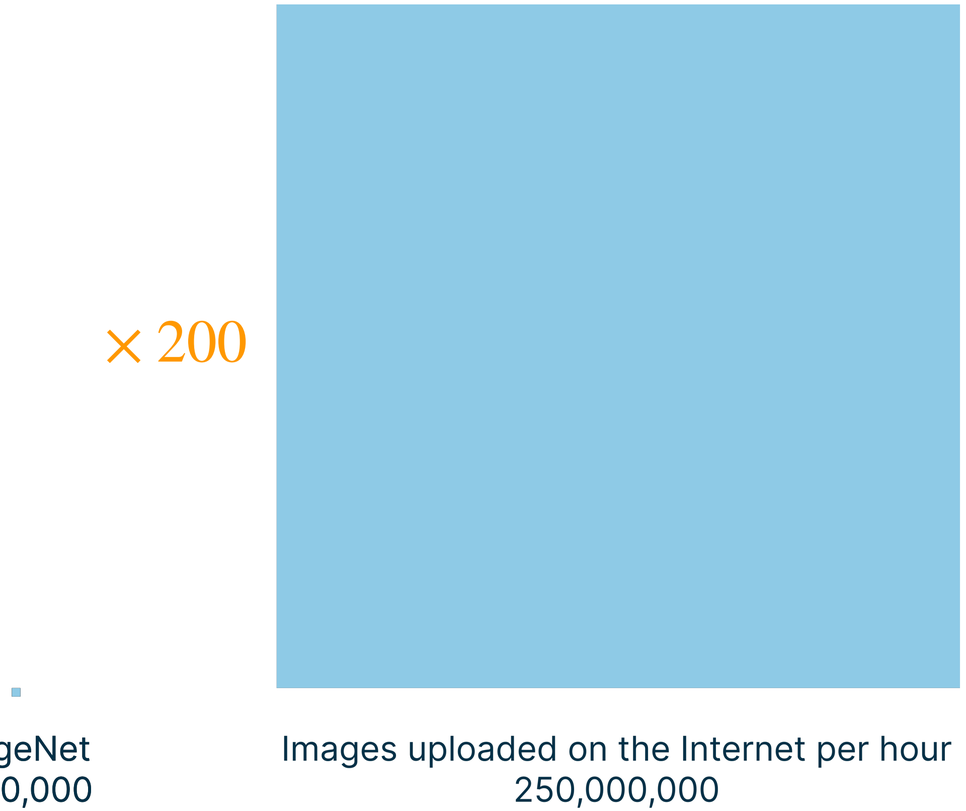

Why self-supervised learning?

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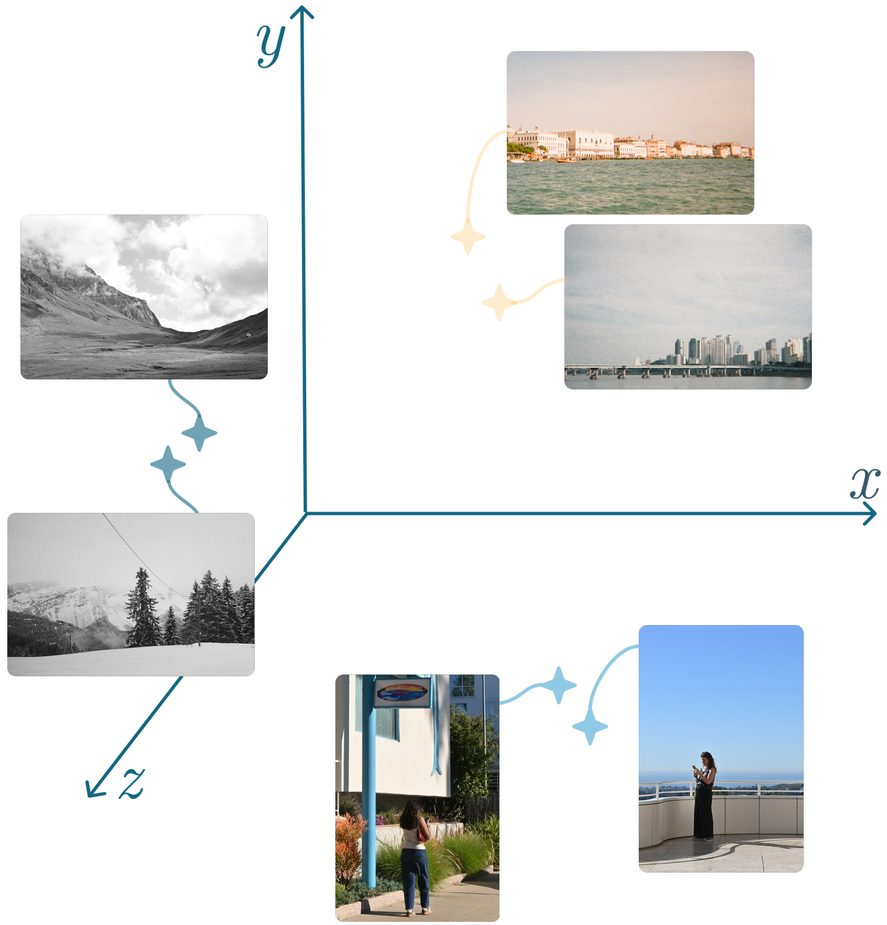
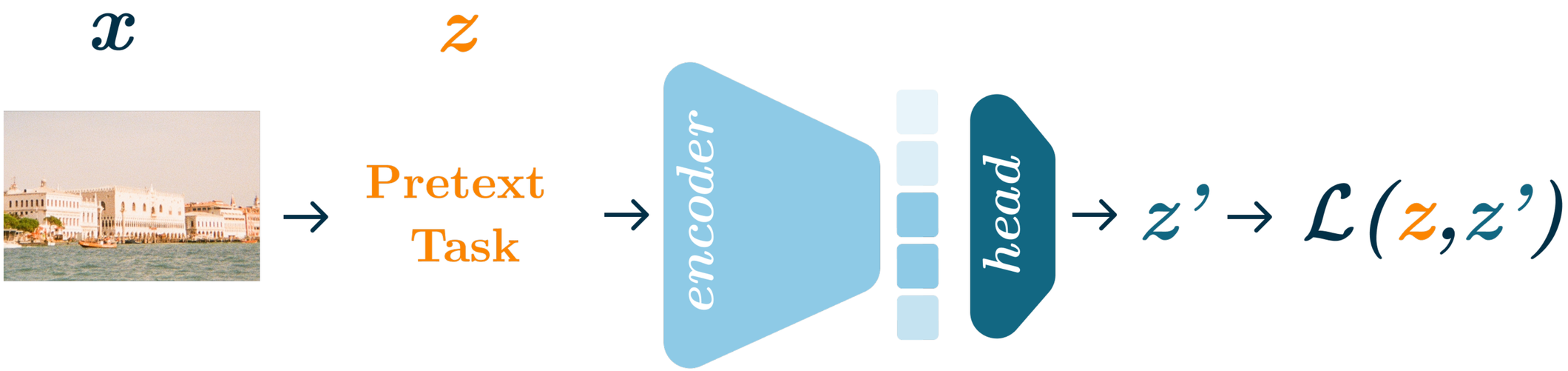
Representation Learning:
Strong representations transfer well



Self-supervised representations

leveraging supervision from data itself

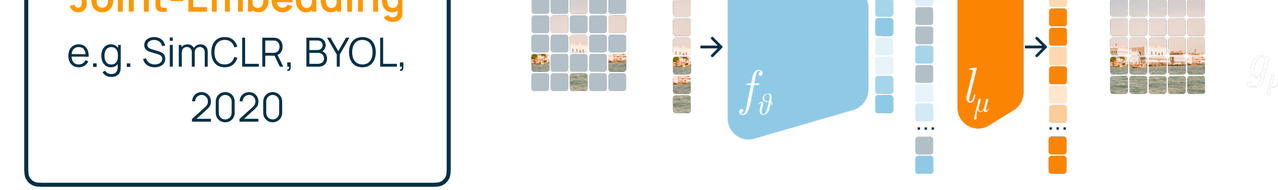


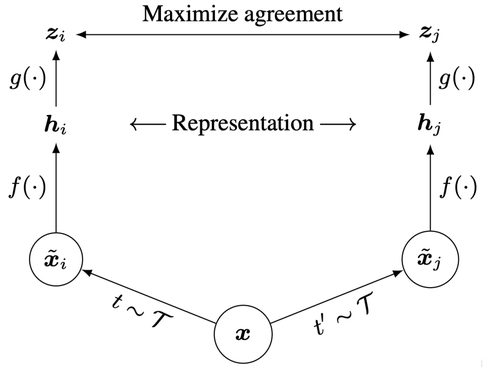
Self-supervised representations

leveraging supervision from data itself

Handcrafted Tasks
e.g. colorisation, 2016

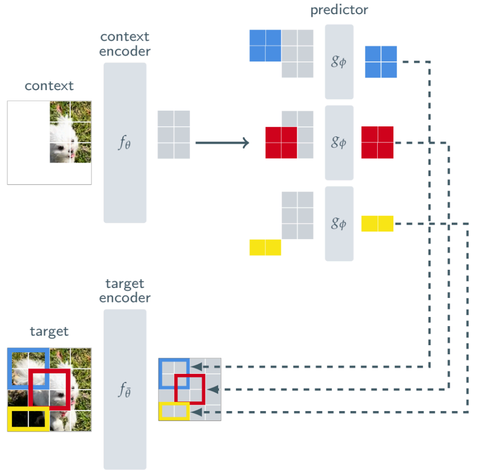
Joint-Embedding
e.g. SimCLR, BYOL, 2020





Masked Image Modelling
e.g. MAE, 2021

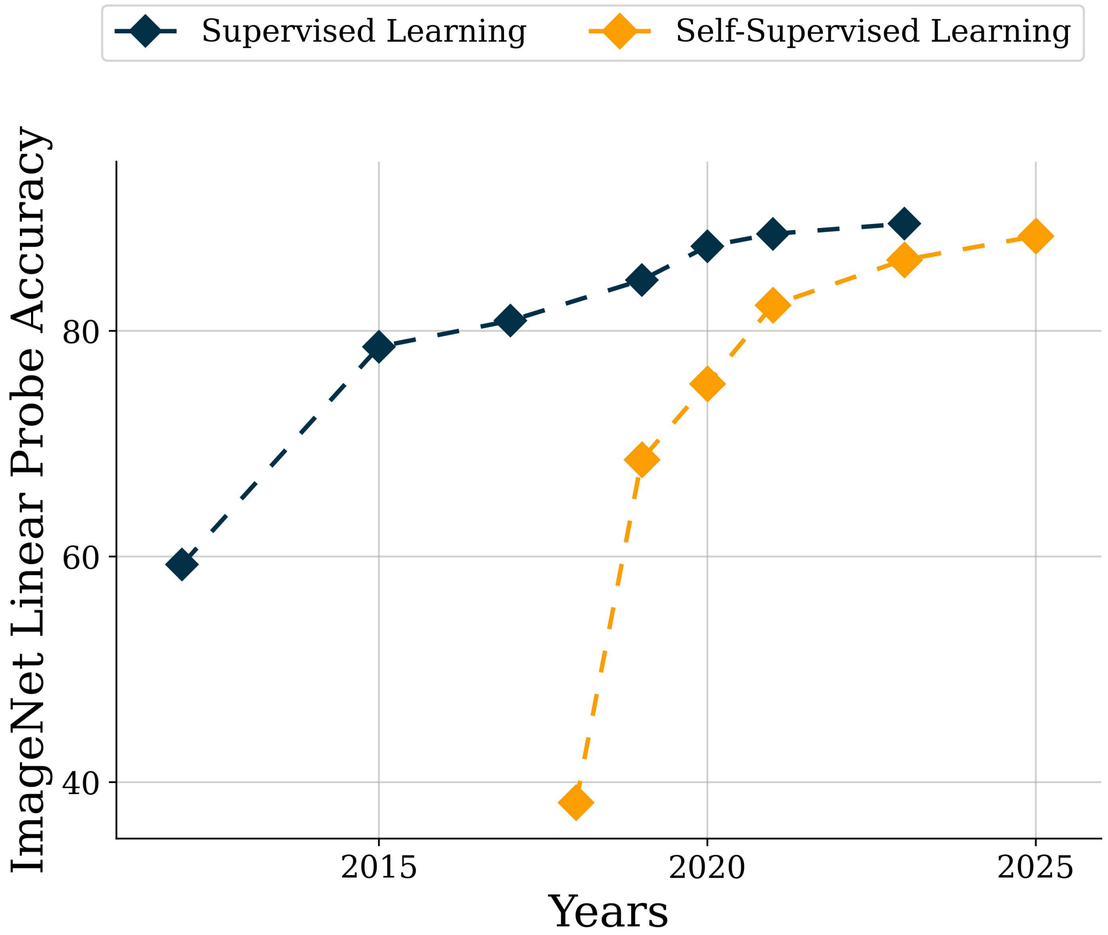
Merging Objectives & Scaling
e.g. DINOv2, I-JEPA, 2023



Zhang et al, ECCV 2016 • Chen et al, ICML 2020 • Grill et al, 2020 · He et al, CVPR 2022 • Assran et al, CVPR 2023

Self-supervised representations

Why are these representations effective?

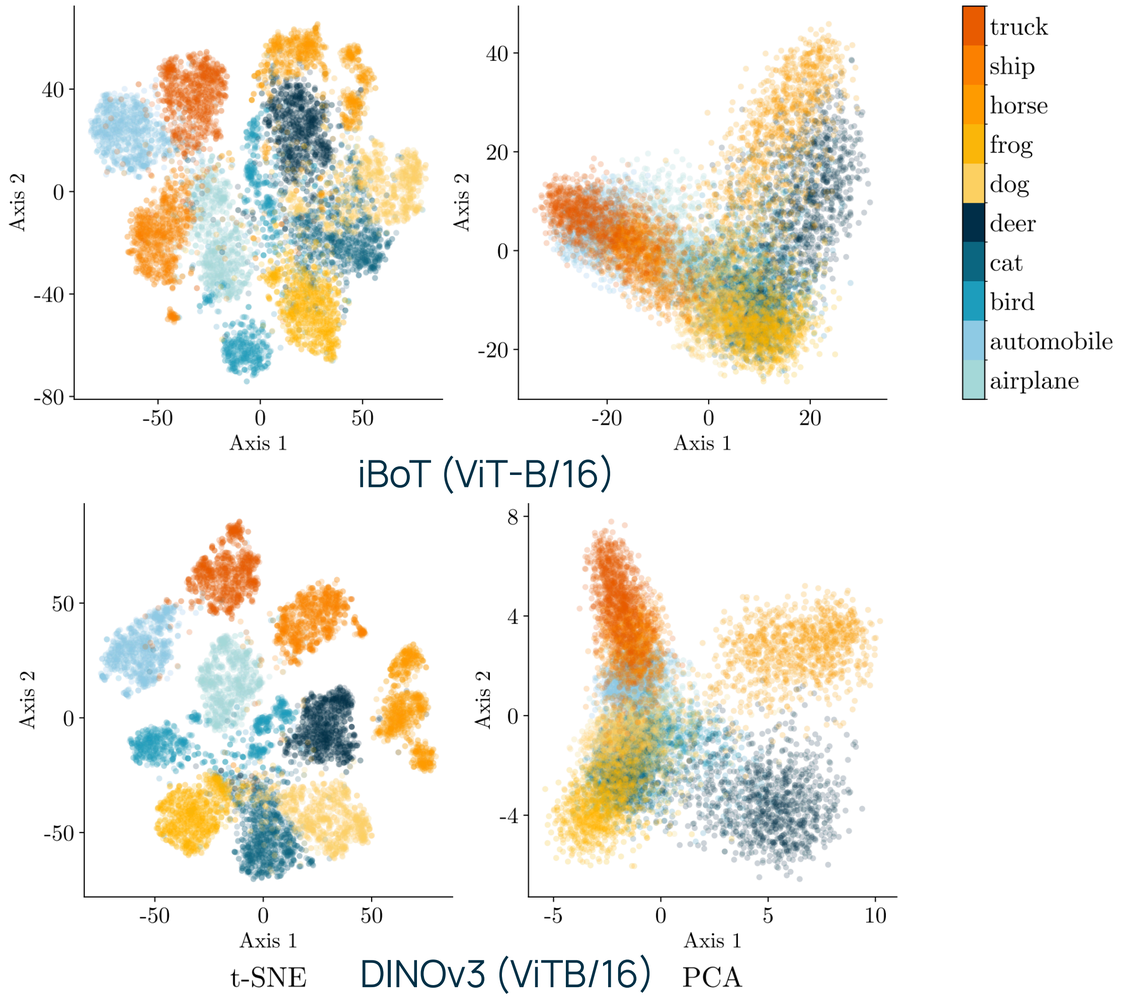


DINOv3, Siméoni et al, 2025

Why capture geometry?

Answers can lie in the **Geometry of representations**:
properties, measurement, and relationships of representations in space

Why capture geometry?



Visualizing CIFAR-10 Representations from Self-Supervised Models via Dimensionality Reductions

iBot, Zhou et al, ICLR 2022 · DINOv3, Siméoni et al, 2025

Why capture geometry?

During Training		*After Training*
Informing the design of training objectives	Diagnosing failure modes Understanding what networks learn	Enabling unsupervised model evaluation

Thesis Objectives

Objective I

Shaping Representations

Preventing known failure modes during training

CAPTURING GEOMETRY

via minimum spanning trees

Objective II

Evaluating Representations

Studying quality beyond downstream performances

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Minimum Spanning Trees

Connecting the dots to capture dimensionality

PART I

Shaping Representations
Against Dimensional Collapse

Enforcing uniformity to bypass collapse in joint-embedding self-supervised learning

T-REGS: Minimum Spanning Regularization for Self-Supervised Learning
J.M., D. Loiseaux, V. Kalogeiton, S. Oudot
NeurIPS 2025

ADAPT: Multimodal Learning for Detecting Stress Under Missing Modalities
J.M., L. Milecki, M. Vakalopoulou, S. Oudot, V. Kalogeiton
MIDL 2024 & CVPR-W 2024

PART II

Evaluating Representations
Beyond Downstream Performances

Studying representations via intrinsic dimension

IdEst: Assessing Self-Supervised Learning Representations via Intrinsic Dimension
J.M., V. Kalogeiton, S. Oudot
ICML 2026

Advisors & Collaborators

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Background

Minimum Spanning Trees

Connecting the dots to capture dimensionality

Minimum Spanning Trees

Definition:

$\mathrm{MST}(P)$: a connected acyclic graph T, spanning P (with vertex set P), whose total length $E(G) := \sum_{(z,z')\in T} ||z - z'||_2$ is minimal among spanning trees of P

Why Minimum Spanning Trees?

Well-known algorithms

Kruskal's algorithm ($\mathcal{O}(n^2 \cdot \log n)$)

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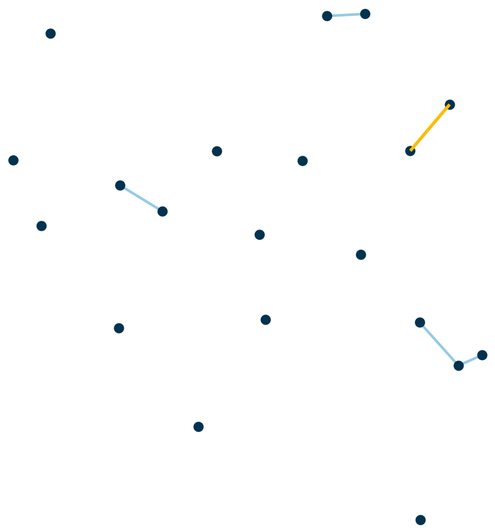
Well-known algorithms



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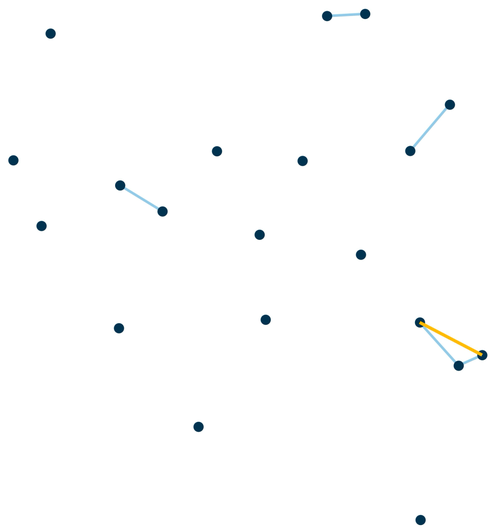


Kruskal's algorithm ($\mathcal{O}(n^2 \cdot \log n)$)

Why Minimum Spanning Trees?

Well-known algorithms

```python
from scipy.spatial.distance import pdist, squareform
from scipy.sparse.csgraph import minimum_spanning_tree

dists = squareform(pdist(z))  # pairwise distances
mst = minimum_spanning_tree(dists)
length =  mst.sum()
```

MST length via **scipy**

```python
from ripser import ripser

pd = ripser(z, maxdim=0)["dgms"][0]
# remove the infinite bar (last connected component)
finite = pd[pd[:, 1] != np.inf]
length = (finite[:, 1] - finite[:, 0]).sum()
```

MST length via **ripser**

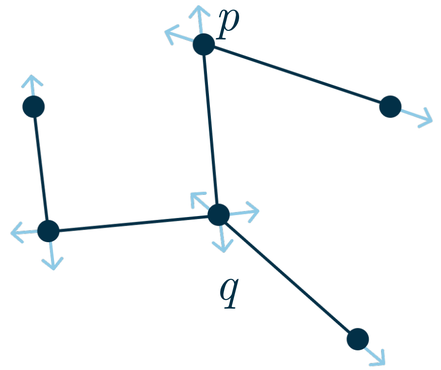
Why Minimum Spanning Trees?

Well-known algorithms

MST's length is differentiable a.e

For fixed n and P a sufficiently regular metric space measure, MST length is Lipschitz stable as a result is differentiable a.e. (generalized Rademacher's theorem):

$$(\nabla_P \, | \, E(\mathrm{MST}(P)) \, | \,)_P = \sum_{(p,q) \text{ edge} \in \mathrm{MST}(P)} \frac{(p-q)}{||p-q||_2}$$



Differentiability of Lipschitz Functions on Metric Measure Space, Cheeger, 1999 · Optimizing persistent homology-based functions, Carriere et al, ICML 2021
A Framework for Differential Calculus on Persistence Barcodes, Leygonie et al, FoCM 2021

Why Minimum Spanning Trees?

Well-known algorithms

MST's length is differentiable a.e.

Captures intrinsic properties

Theorem [Steele, 1988]: Let X_n be an i.i.d n-sample of a probability measure P_X with compact support on $\mathbb{R}^d$. For $d \geq 2$, the growth rate of the length of the $MST(X_n)$ is controlled as follows:

$$L(\mathrm{MST}(X_n)) \sim Cn^{(d-1)/d} \ \ almost\ surely\ , as\ n \to \infty$$

where $\sim$ denotes asymptotic convergence, and where C is a constant depending only on P_X and d.

Growth Rates of Euclidean Minimal Spanning Trees with Power Weighted Edges, Steele, 1988

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Asymptotically

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Asymptotically, the growth rate of the MST's length

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$$L(\mathrm{MST}(X_n)) \sim Cn^{(d-1)/d} \ \ almost\ surely\ , as\ n \to \infty$$

where $\sim$ denotes asymptotic convergence, and where C is a constant depending only on P_X and d.

Asymptotically, the growth rate of the MST's length is governed by the **dimension** d

Differentiability of Lipschitz Functions on Metric Measure Space, Cheeger, 1999
Optimizing persistent homology-based functions, Carriere et al, ICML 2021

Why Minimum Spanning Trees?

Well-known algorithms

MST's length is differentiable a.e.

Captures intrinsic properties

Theorem [Costa and Hero, 2006]: Let X_n be an iid n-sample of a probability measure on $\mathscr{M}$ with density f w.r.t. μ. Then, there exists a constant independent of f and of $\mathscr{M}$ such that:

$$n^{-(d-1)/d} \cdot L(\mathrm{MST}(X_n)) \xrightarrow[n\to\infty]{} C \int f^{\frac{d-1}{d}} \mathrm{d}\mathscr{H}$$

Where C does not depend neither on f nor on $\mathscr{M}$

Refined to the **Riemannian** manifold setting, making the dependence of the limit on the distribution f explicit, linking to its intrinsic dimension d.

Why Minimum Spanning Trees?

Well-known algorithms

MST's length is differentiable a.e.

Captures intrinsic properties

Algorithm 1 Computation of $\dim_{\mathrm{MST}}$

Input: The set of representations $Z = \{z_1, \ldots, z_N\}$, minimun sample size $n_{\min}$, skip size Δ
Output: The estimated dimension: $\dim_{\mathrm{MST}}(Z)$
Initialize $n \leftarrow n_{\min}$, $E \leftarrow []$
while $n < N$ **do**
  $Z_n \leftarrow \mathrm{sample}(Z, n)$ # random sampling of n points
  $E[i] \leftarrow L\left(\mathrm{MST}(Z_n)\right)$ # computation of the minimum spanning tree
  $n \leftarrow n + \Delta$
end while
$m, b \leftarrow$ linear regression$(\log(n_{\min} : \Delta : N), \log(E))$ # linear regression of the log-log plot
$\dim_{\mathrm{MST}}(\mathrm{Z}) \leftarrow 1/(1 - m)$

Intrinsic dimension, persistent homology and generalization in neural networks, Birdal et al, NeurIPS
Fractal dimension and the persistent homology of random geometric complexes, Schweinhart, Advances in Mathematics, 2018

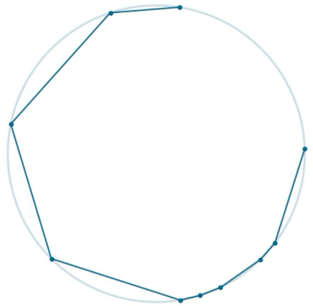
Why Minimum Spanning Trees?

Well-known algorithms

MST's length is differentiable a.e.

Captures intrinsic properties

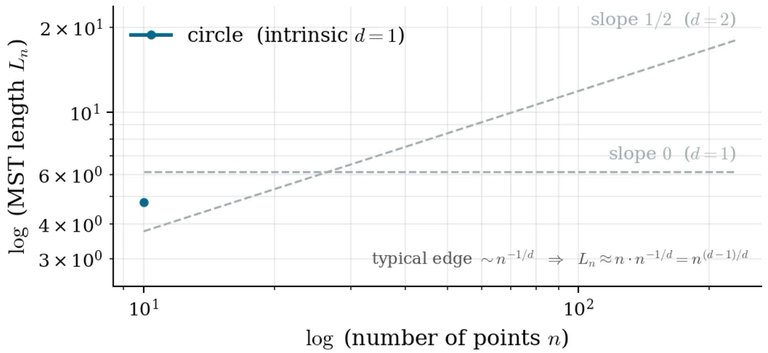
1-D support: points on a circle

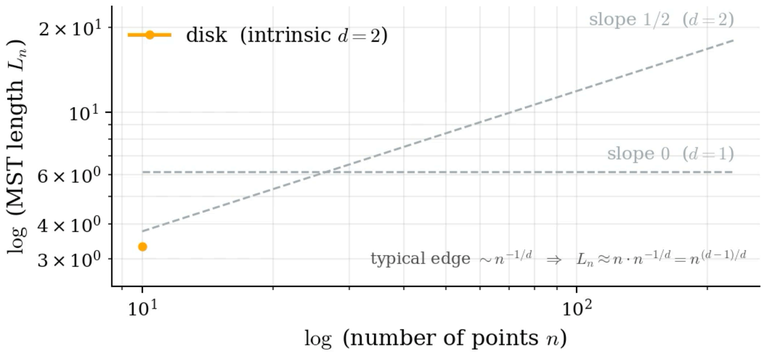


$n = 10 \quad L_n = 4.76 \quad \hat{d} = ?$

2-D support: points in a disk

$n = 10 \quad L_n = 3.32 \quad \hat{d} = ?$





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Part I

Shaping Representations Against Dimensional Collapse

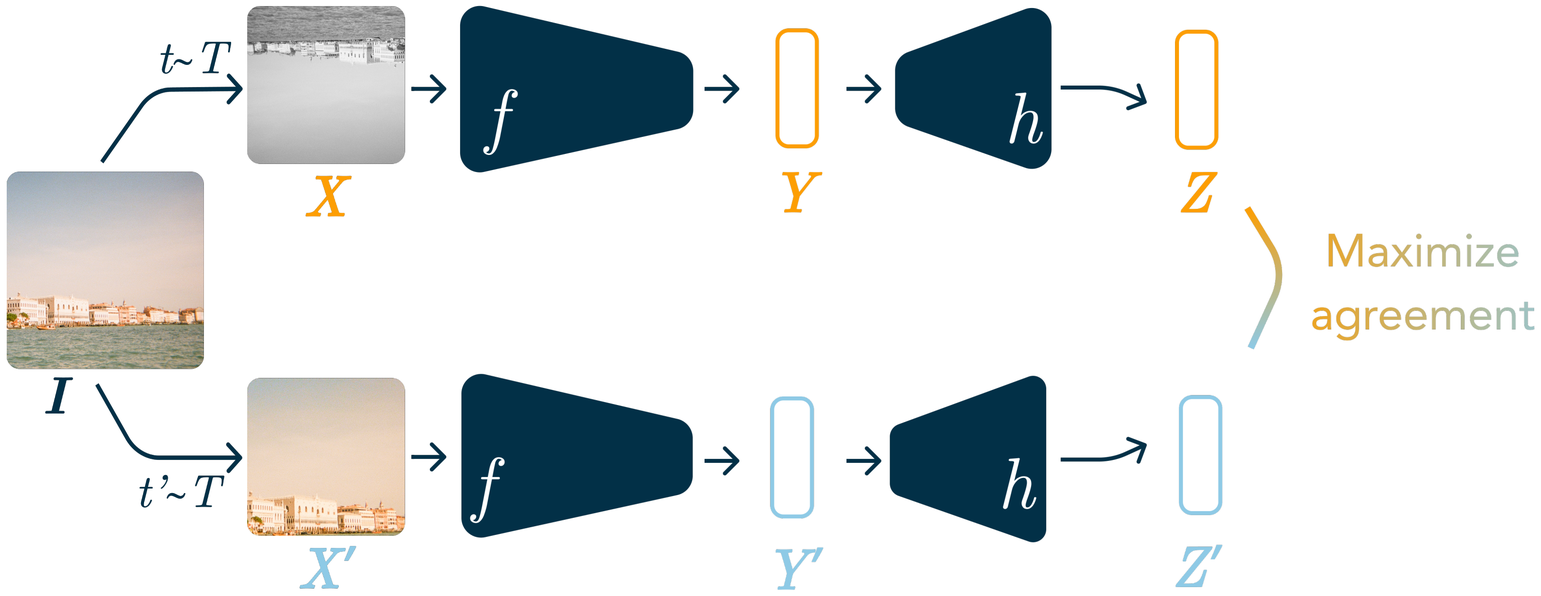
Enforcing uniformity to bypass collapse in joint-embedding self-supervised learning

T-REGS

Enforce uniformity to bypass collapse in joint-embedding architectures through a principle and simple regularization

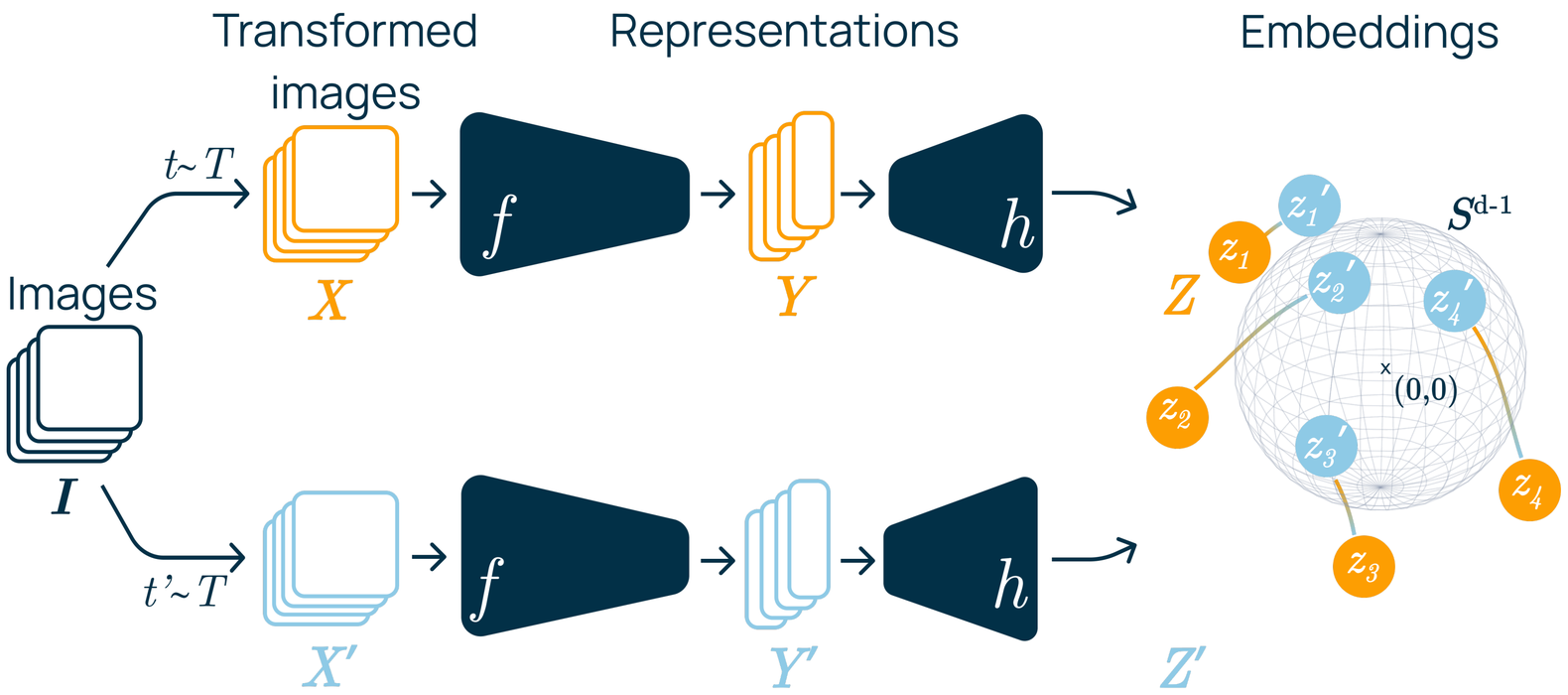
Publication:
T-REGS: Minimum Spanning Tree Regularization for Self-Supervised Learning
J.M., David Loiseaux, Vicky Kalogeiton, Steve Oudot
NeurIPS 2025 (Spotlight)

Collapse or Not? Joint-Embedding SSL



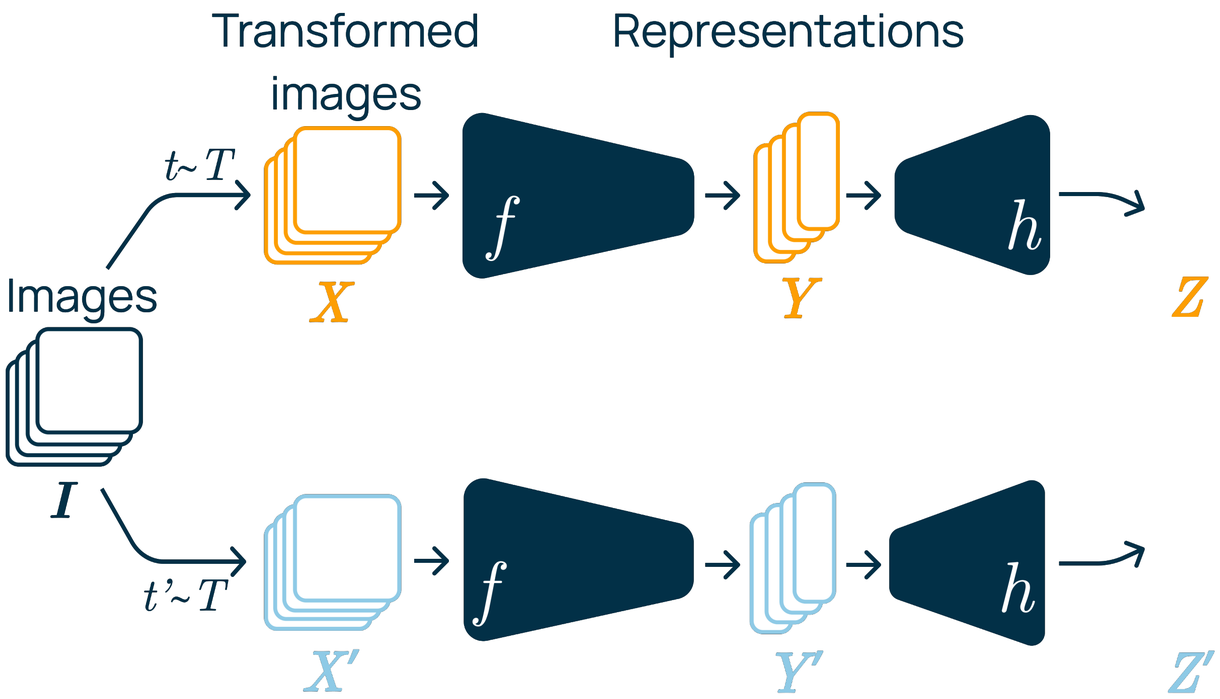
Enforcing semantic invariance

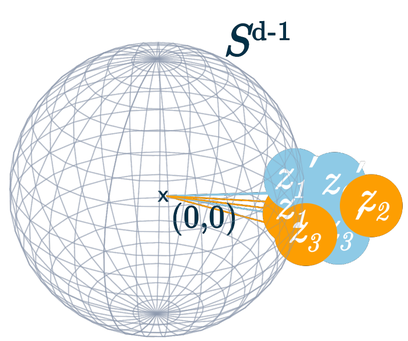
Collapse or Not? Joint-Embedding SSL



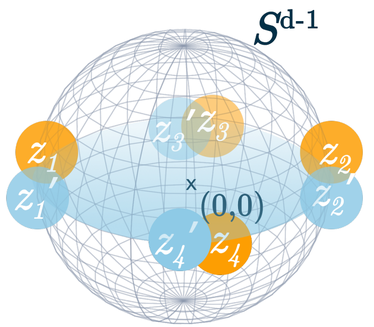
$$\mathscr{L} := \mathscr{L}_{\text{inv}}(Z, Z')$$

Collapse or Not? Joint-Embedding SSL

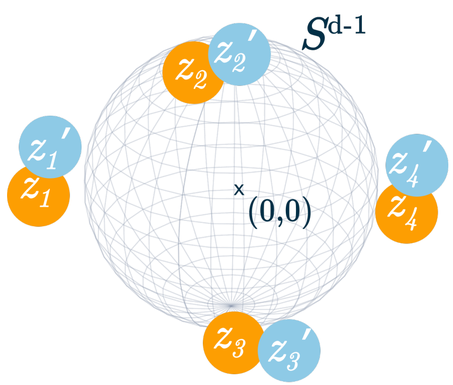




✗ Full Collapse
embeddings collapse to a single point



✗ Dimensional Collapse
embeddings occupy a lower dimension subspace



✓ Uniformly distributed & no collapse

$\mathscr{L} := \mathscr{L}_{\text{inv}}(Z, Z')$ + **anti-collapse mechanism**

Understanding Contrastive Representation Learning through Alignment and uniformity on the Hypersphere, Wang and Isola, ICML 2020
Rethinking uniformity in self-supervised learning, Fang et al, ICLR 2024

Collapse or Not? Joint-Embedding SSL

Method Family	Anti-Collapse Mechanism	Limitations

Collapse or Not? Joint-Embedding SSL

Method Family	Anti-Collapse Mechanism	Limitations
Contrastive (e.g., MOCO, SimCLR)	Explicit negative repulsion on hypersphere	Huge batch/queue sizes, high memory

MOCO, He et al, CVPR 2020 · SimCLR, Chen et al, ICML 2020

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Contrastive (e.g., MOCO, SimCLR)	Explicit negative repulsion on hypersphere	Huge batch/queue sizes, high memory
Self-distillation (e.g., BYOL, DINO)	**Asymmetry (stop-grad, EMA teacher, centering)**	**No formal loss barrier; sensitive to hyperparameter balance**

DINO, Caron et al, CVPR 2021 · BYOL, Grill et al, NeurIPS 2020

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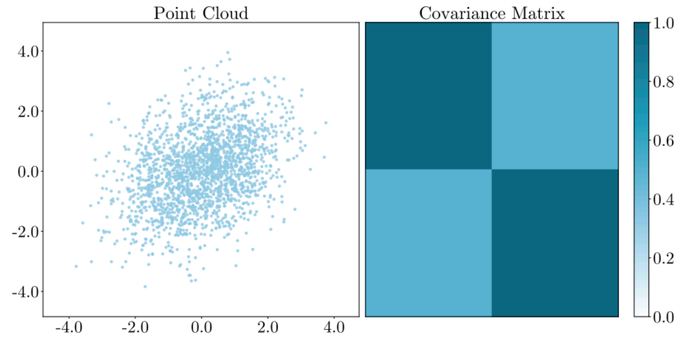
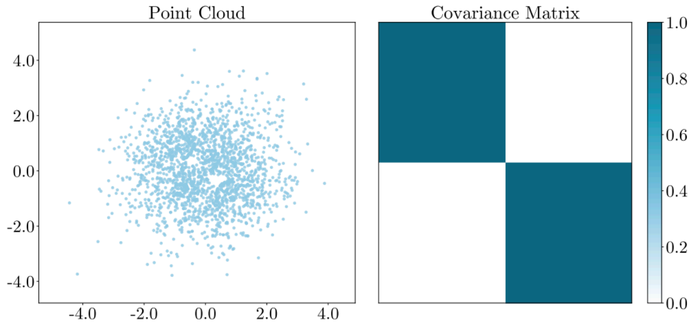
Redundancy reduction (e.g. VICReg, Barlow Twins)	$\mathscr{L}_{\text{reg}}(Z) = \|\|\text{cov}(Z) - I\|\|_F^2$	**Controls up to 2nd-order statistics**



Initial Point Cloud:
sampling of a non-isotropic Gaussian measure



After Optimization
with $\mathscr{L}_{\textbf{reg}}(Z) = ||\,\textbf{cov}(Z) - I\,||_F^2$

Barlow Twins, Zbontar et al, ICML 2021 · VICReg, Bardes et al, ICLR 2022

Collapse or Not? Joint-Embedding SSL

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Redundancy reduction (e.g. VICReg, BarlowTwins)	$\mathscr{L}_{\text{reg}}(Z) = \lvert\lvert \text{cov}(Z) - I \rvert\rvert_F^2$	Controls up to 2nd-order statistics
Uniformity (e.g. Gaussian pairwise potential, Wasserstein distance)	**Matches global uniform hyperspherical prior**	**Dense pairwise gradients, slow computation**

Gaussian Pairwise potential, $\mathscr{L}_{\text{reg}}(Z) = \log \mathbb{E}_{x,y \sim p_d}[e^{-t\lvert\lvert f(x)-f(y)\rvert\rvert_2^2}]\,, t > 0$

Wasserstein distance, $\mathscr{L}_{\text{reg}}(Z) := \mathscr{W}_2^2(\mu_z, \mathscr{U}_{\mathbb{S}^{d-1}}) \approx \mathscr{W}_2^2(\mathscr{N}(\bar{Z}, \text{cov}(Z)), \mathscr{N}(0, I/d))$

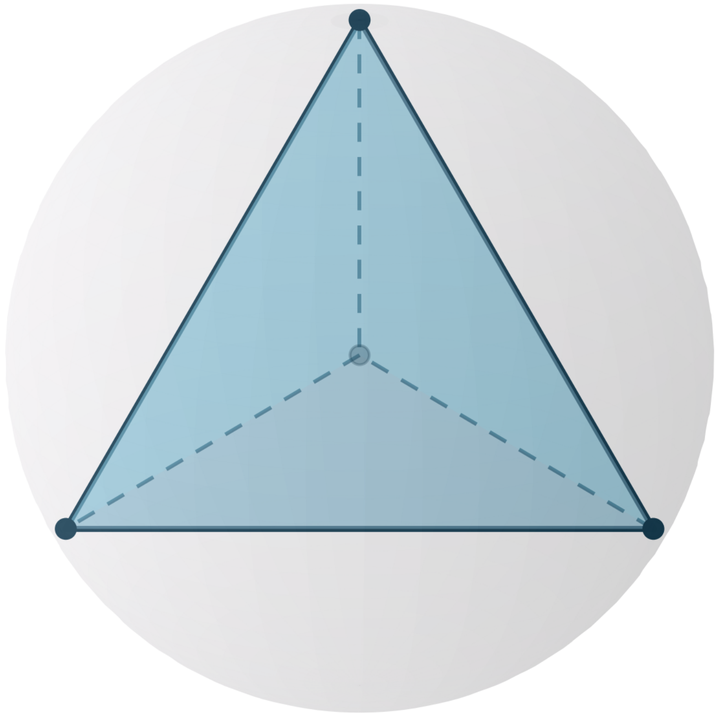
Rethinking uniformity in SSL, Wang et al, ICLR 2024
Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere, Wang et al, ICML 2020

Connecting JE-SSL to Tammes Problem

Tammes problem:

Given $n \geq d+1$, which configurations of n points on the sphere $\mathbb{S}^{d-1}$ maximises the minimum/ average distances?

Solution when $n = d+1$



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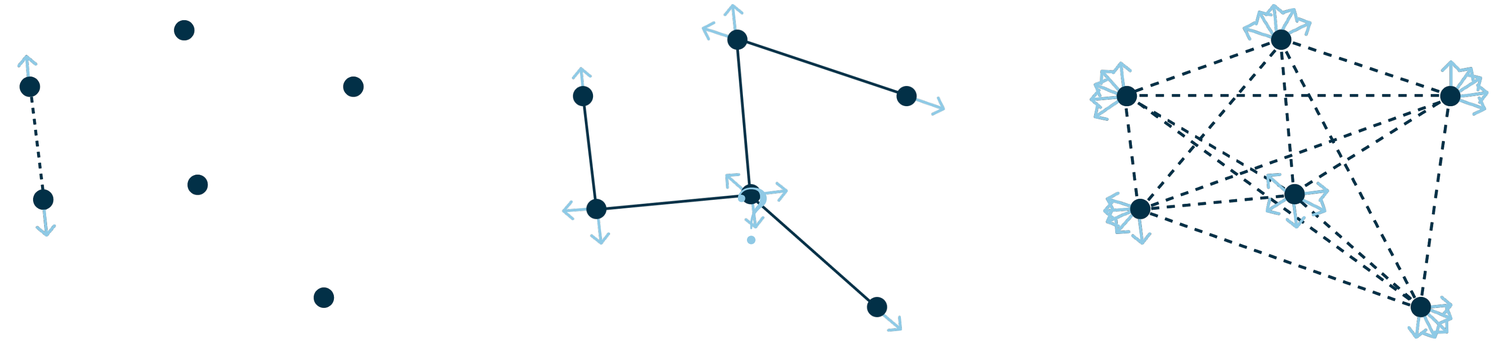

$P \to \min_{p \neq q \in P} ||p-q||_2$

$P \to \sum_{(p,q) \in \mathrm{MST}(P)} ||p-q||_2$

$P \to \sum_{p,q \in P} ||p-q||_2$

sparse gradients

dense gradients

T-REGS

Increase the length of the MST constrained to the sphere

1. Theoretical guarantees: MST and sample uniformity
2. Regularization term: $-\gamma \frac{1}{n} E(\mathrm{MST}(Z)) + \lambda \frac{1}{n} \sum_i (\|z_i\|_2 - 1)^2$
3. Effectiveness on standard SSL benchmark

T-REGS

MST & Uniformity: asymptotic regime ($n \to \infty$)

$\mathscr{M}$, a compact Riemannian $(d-1)$-manifold with $(d-1)$-dimensional Hausdorff measure $\mathscr{H}$

Theorem [Costa and Hero, 2006]: Let X_n be an iid n-sample of a probability measure on $\mathscr{M}$ with density f w.r.t. μ. Then, there exists a constant independent of f and of $\mathscr{M}$ such that:

$$n^{(d-1)/d} \cdot E(\mathrm{MST}(X_n)) \underset{n\to\infty}{\longrightarrow} C \int f^{\frac{d-1}{d}} \mathrm{d}\mathscr{H} \approx \exp H^{\mathrm{S}^{d-1}}_{1-\frac{1}{d-1}}(f)$$

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Determining Intrinsic Dimension and Entropy of High-Dimensional Shape Spaces. Costa et al, 2006
Maximum entropy autoregressive conditional heteroskedasticity model. Park et al, 2009

T-REGS

MST & Uniformity: asymptotic regime ($n \to \infty$)

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On a compact Riemannian manifold asymptotically,
the limiting MST's length

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On a compact Riemannian manifold asymptotically,
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which itself is maximised at the **uniform distribution**

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T-REGS

MST & Uniformity: sparse regime ($n \approx d$)

$n \leq d + 1$, e.g. batch sizes are often smaller than or comparable to the ambient dimension ; assume Z lies in a closed Euclidean d-ball B of radius r centered at the origin.

Theorem [M., Loiseaux, Kalogeiton, Oudot, 2025]: Under the above conditions, the maximum of the $E(\mathrm{MST}(Z))$, over the point sets $X \subset B$ of fixed cardinality n is attained when the points of X lie on the sphere $S = \partial B$, at the vertices of a regular $(n-1)$-simplex that has S as its smallest circumscribing sphere.

When the number of points is closer to the ambient space

T-REGS

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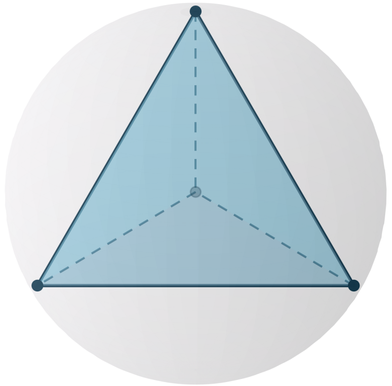

Regular 3-simplex: tetrahedron

T-REGS

A regularization method

Given, $Z = \{z_0, \ldots, z_n\} \in \mathbb{R}^d$, $\mathscr{L}(Z) = -\gamma \frac{1}{n} E(\mathrm{MST}(Z)) + \lambda \frac{1}{n} \sum_i (\|z_i\|_2 - 1)^2$

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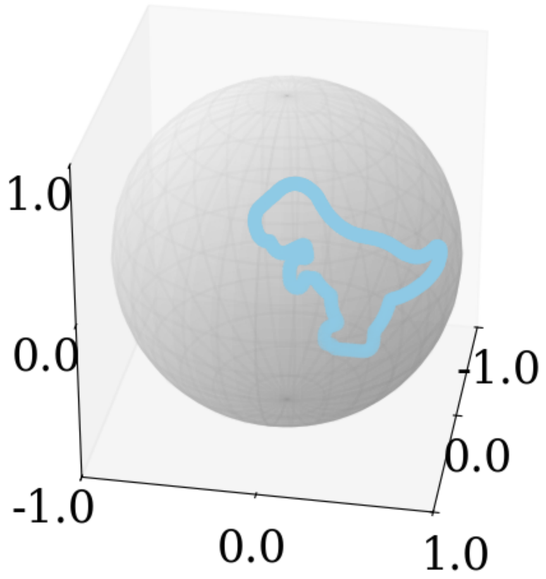

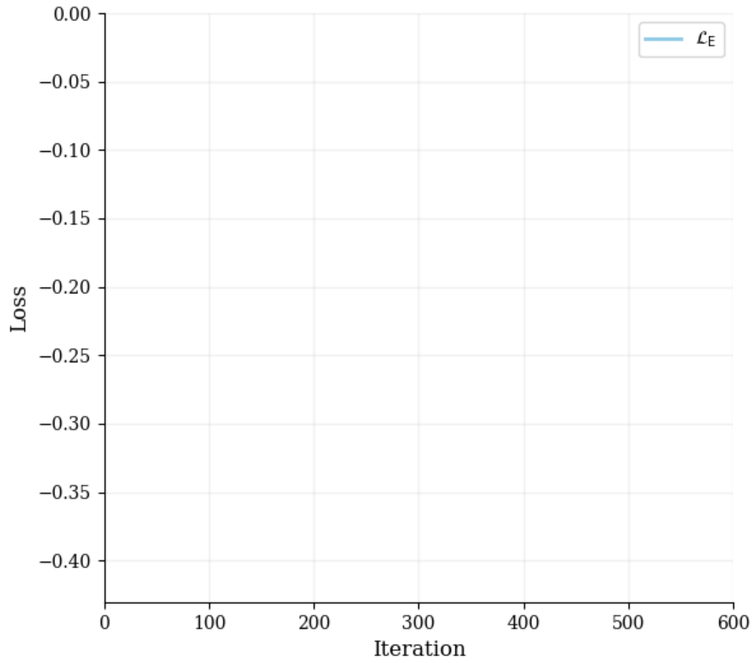

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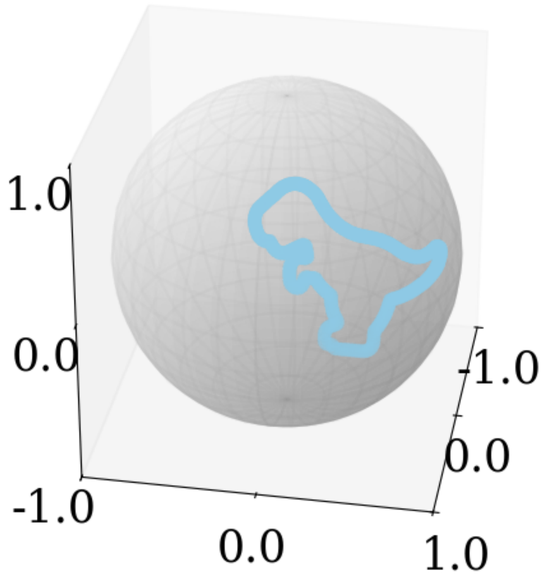
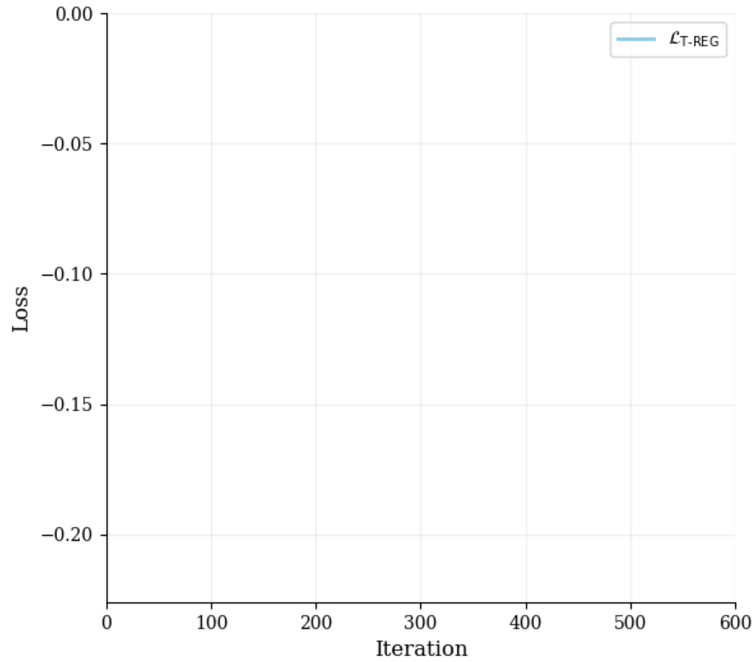
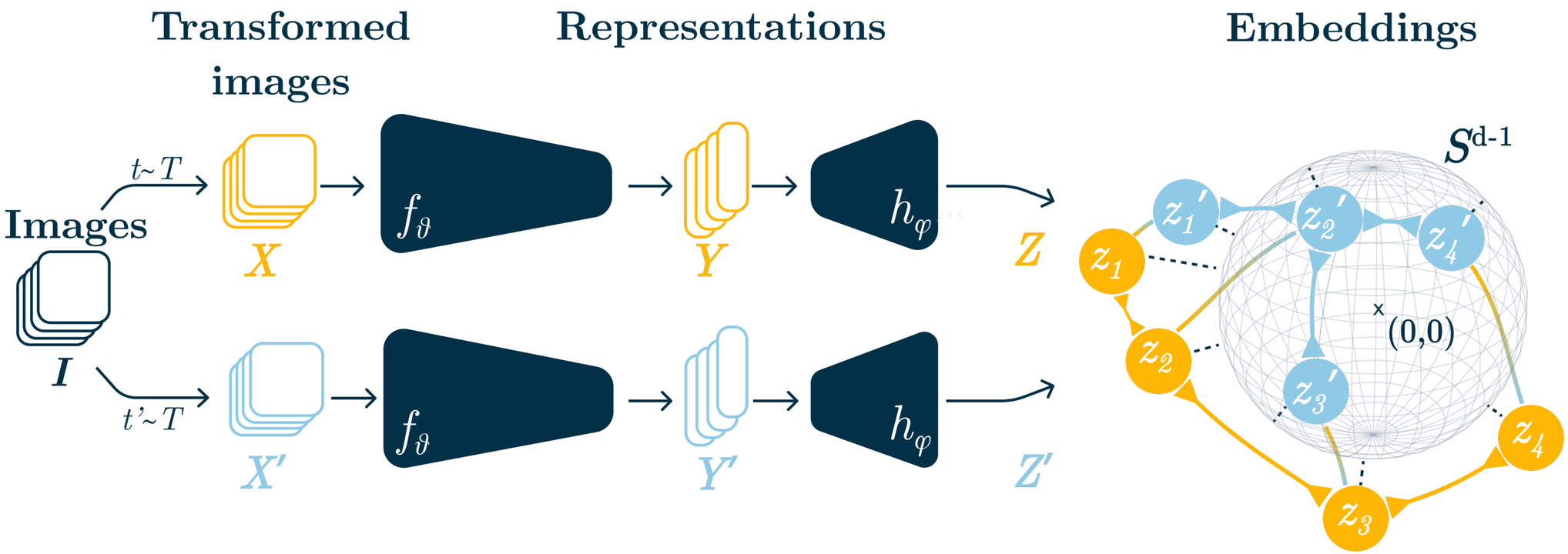


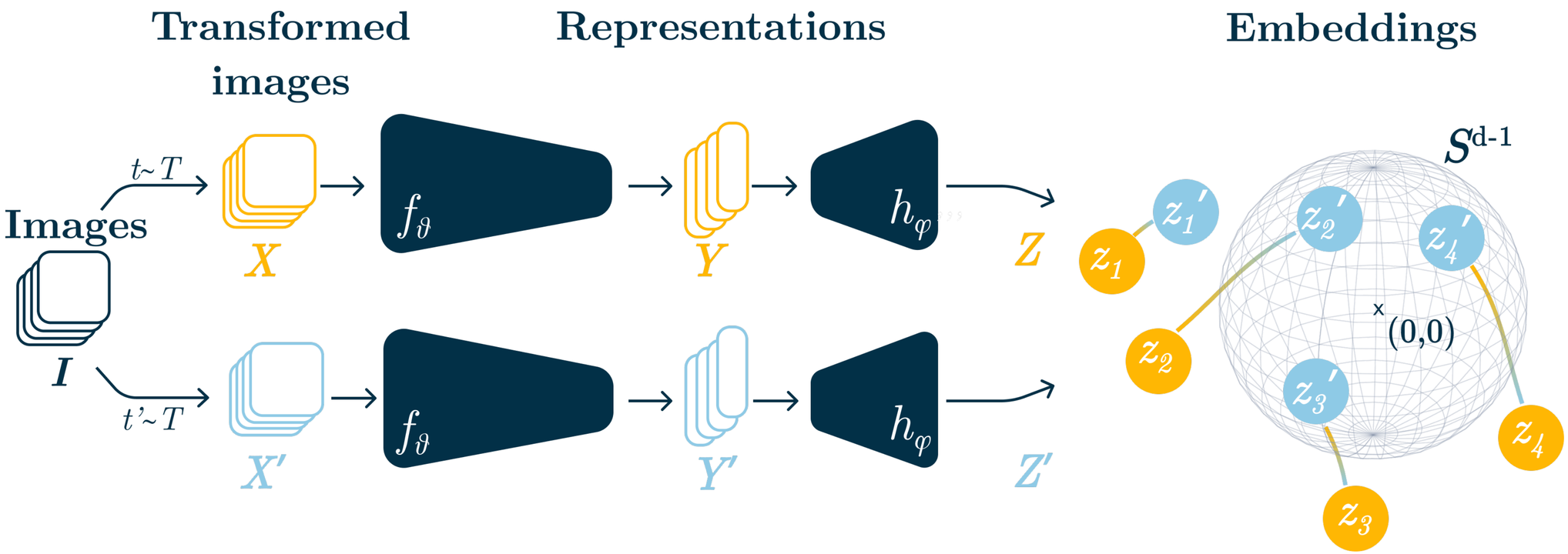


T-REGS



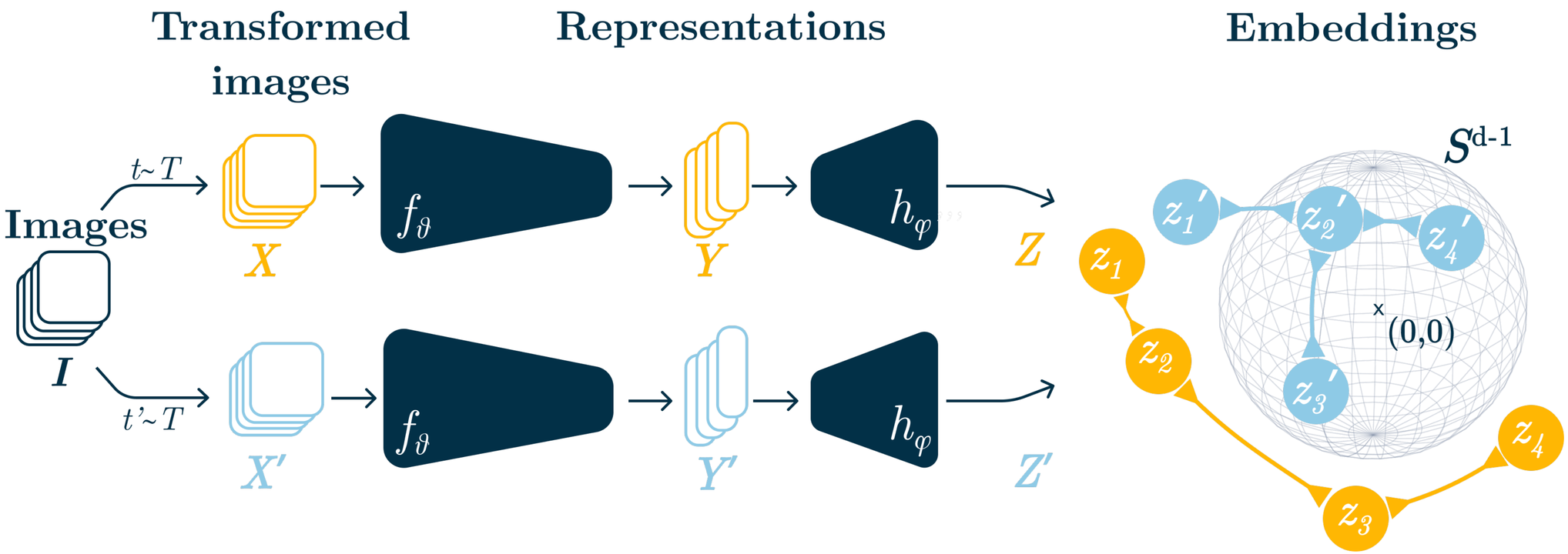
$$\mathscr{L}(Z, Z') = \beta\,\mathscr{L}_{\mathrm{SSL}}(Z, Z') + \gamma\,\mathscr{L}_{\mathrm{E}}(Z) + \lambda\,\mathscr{L}_{\mathrm{S}}(Z) + \gamma\,\mathscr{L}_{\mathrm{E}}(Z') + \lambda\,\mathscr{L}_{\mathrm{S}}(Z')$$

T-REGS



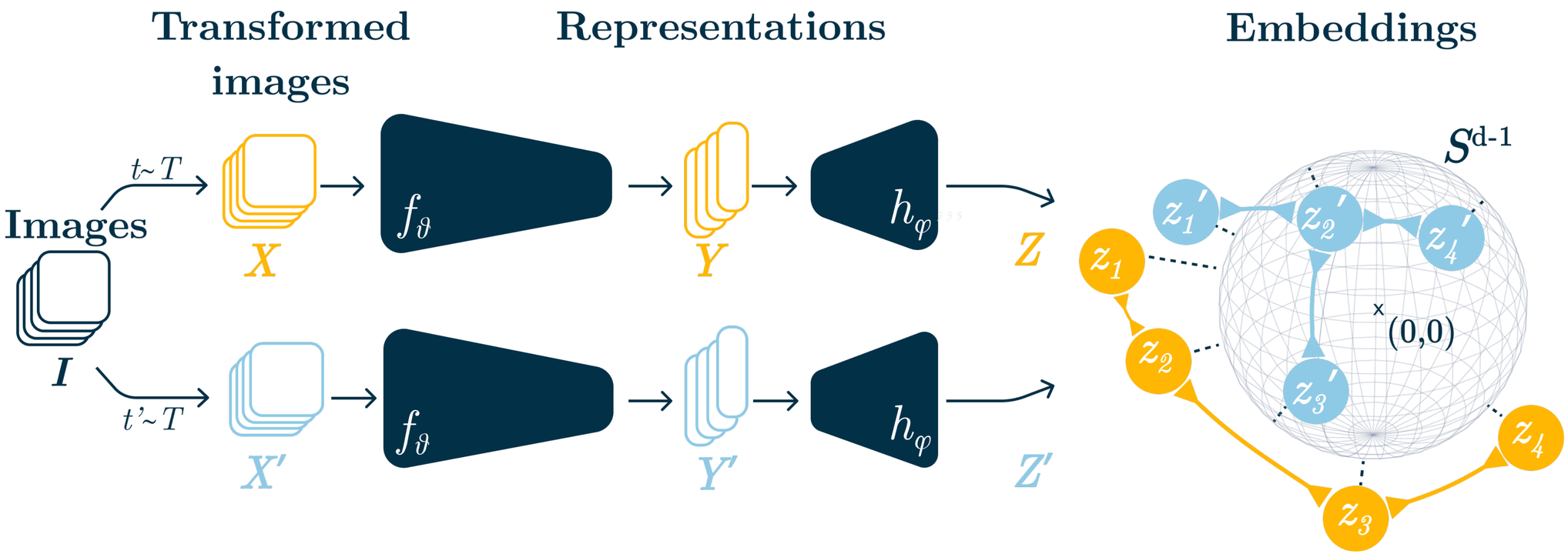
$$\mathscr{L}_{\mathrm{SSL}}(Z, Z') \text{ or } \mathscr{L}_{\mathrm{MSE}}(Z, Z')$$

T-REGS



$L(\mathrm{MST}(Z))$ and $L(\mathrm{MST}(Z'))$

T-REGS



$\mathscr{L}_S(Z)$ and $\mathscr{L}_S(Z')$

T-REGS

Evaluation on standard SSL benchmark

Method		CIFAR-10	CIFAR-100
Zero-CL		91.3	**68.5**
	$+\mathscr{L}_u$	91.3	68.4
	$+\mathscr{W}_2$	91.4	**68.5**
MoCo v2		90.7	60.3
	$+\mathscr{L}_u$	91.0	61.2
	$+\mathscr{W}_2$	91.4	63.7
BYOL		89.5	63.7
	$+\mathscr{L}_u$	90.1	62.7
	$+\mathscr{W}_2$	90.1	65.2
	$+\mathscr{L}$ T-REGS	90.4	65.7
Barlow Twins		91.2	68.2
	$+\mathscr{L}_u$	91.4	68.4
	$+\mathscr{W}_2$	91.4	**68.5**
	$+\mathscr{L}$ T-REGS	**91.8**	**68.5**
$\mathscr{L}$ MSE	$+\mathscr{L}$ T-REGS	91.3	67.4

Comparison with $\mathscr{W}_2$ regularization on CIFAR-10/100.

T-REGS demonstrate strong standalone performances

T-REGS

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$\mathscr{L}$ MSE	$+\mathscr{L}$ T-REGS	91.3	67.4

Comparison with $\mathscr{W}_2$ regularization on CIFAR-10/100.

T-REGS consistently outperforms both the baselines & variants using additional regularization terms $\mathscr{L}_{\mathrm{U}}$ and $\mathscr{W}_2$

T-REGS

Evaluation on standard SSL benchmark

T-REGS is competitive with methods that use the same number of views.

# views	Method		ImageNet-100	ImageNet-1k	
			Top-1	Batch Size	Top-1
8	SwAV		74.3	4096	66.5
	FroSSL		79.8	-	-
	SSOLE		**82.5**	256	**73.9**
2	SimCLR		77.0	4096	66.5
	MoCo V2		79.3	256	67.4
	SimSiam		78.7	256	68.1
	W-MSE		69.1	512	65.1
	Zero-CL		79.3	1024	**68.9**
	VICReg		79.4	1024	68.3
	CW-RGP		77.0	512	67.1
	INTL		**81.7**	512	**69.5**
	BYOL		80.3	1024	66.5
		$+\mathscr{L}$T-REGS	80.8	1024	67.2
	BarlowTw		80.2	2048	67.7
		$+\mathscr{L}$T-REGS	**80.9**	2048	67.8
	$\mathscr{L}$MSE	$+\mathscr{L}$T-REGS	80.3	512	**68.8**

Comparison on ImageNet100 & ImageNet-1k .

T-REGS

Evaluation on standard SSL benchmark

Method	Complexity	B range	D range	Wall-clock time
SimCLR	$\mathcal{O}(B^2 \cdot D)$	[2048-4096]	[256-1024]	0.22 +/- 0.03
VICReg	$\mathcal{O}(B \cdot D^2)$	[1024-4096]	[4096-8192]	0.23 +/- 0.02
$\mathcal{L}_{\text{MSE}} + \mathcal{L}_{\text{T-REGS}}$	$\mathcal{O}(B^2(D \cdot \log B))$	[512-2048]	[512-2048]	**0.20 +/- 0.001**

Complexity and compute cost on ImageNet-1k.
Comparison between different methods is performed, with training on ImageNet-1k distributed across 4 Tesla H100 GPUs.
The wall-clock time (sec/step) is averaged over 500 steps, with B=512 and D=1024

SimCLR, Chen et al, ICML 2020 · VICReg, Bardes et al, ICLR 2021

From training to evaluating:

Can geometry provide insights into representations' quality?

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Minimum Spanning Trees

Connecting the dots to capture dimensionality

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Against Dimensional Collapse

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PART II

Evaluating Representations
Beyond Downstream Performances

Studying representations via intrinsic dimension

Part II

Evaluating Representations Beyond Downstream performances

Evaluating representations via intrinsic dimension

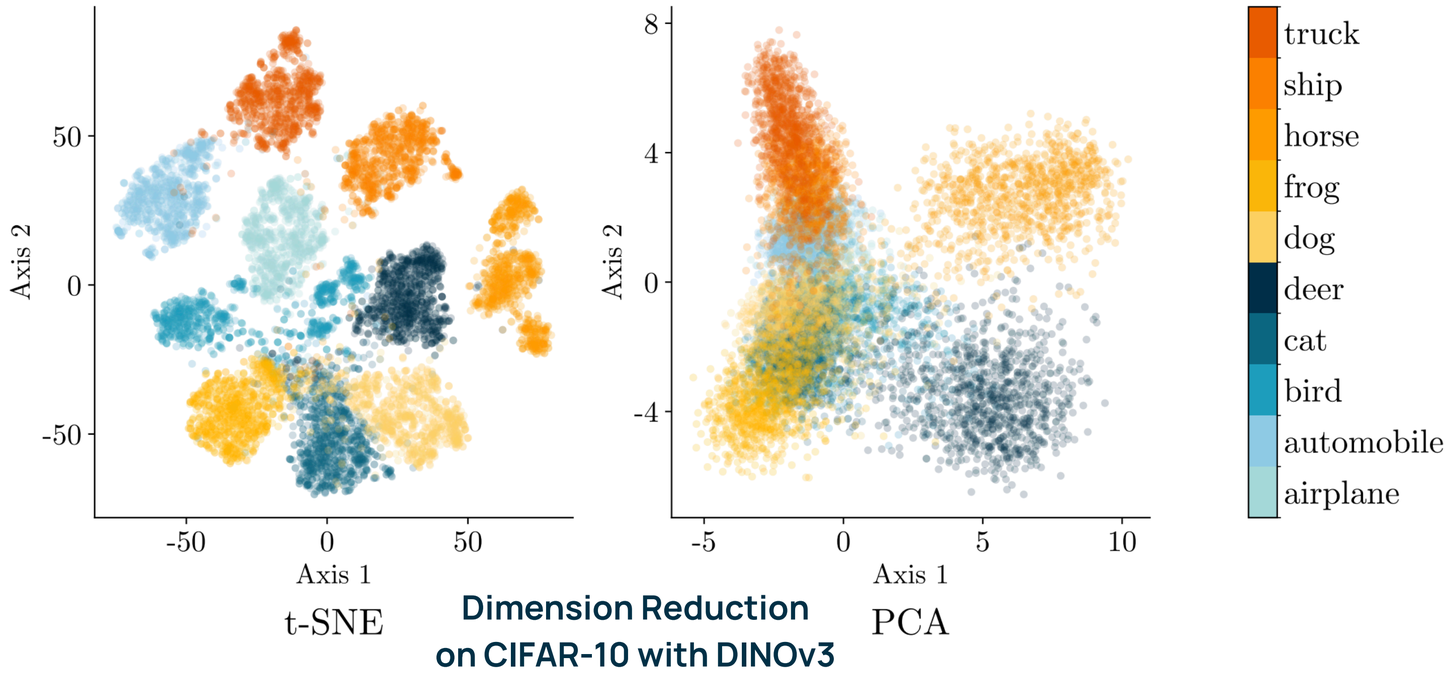
IdEst

Evaluating representations via intrinsic dimension

Publication:
IdEst: Assessing Self-Supervised Representations via Intrinsic Dimension,
J.M., Vicky Kalogeiton, Steve Oudot
ICML 2026

Can geometry provide label-free insights into representations' quality?

Manifold Hypothesis: given data observed in an ambient space $\mathbb{R}^p$, the points are assumed to lie on or near a sub-manifold $\mathscr{M} \subset \mathbb{R}^p$ where $\dim(\mathscr{M}) \ll p$



Dimension Reduction on CIFAR-10 with DINOv3

Some semantic structure emerges, even in 2d.

Can geometry provide label-free insights into representations' quality?

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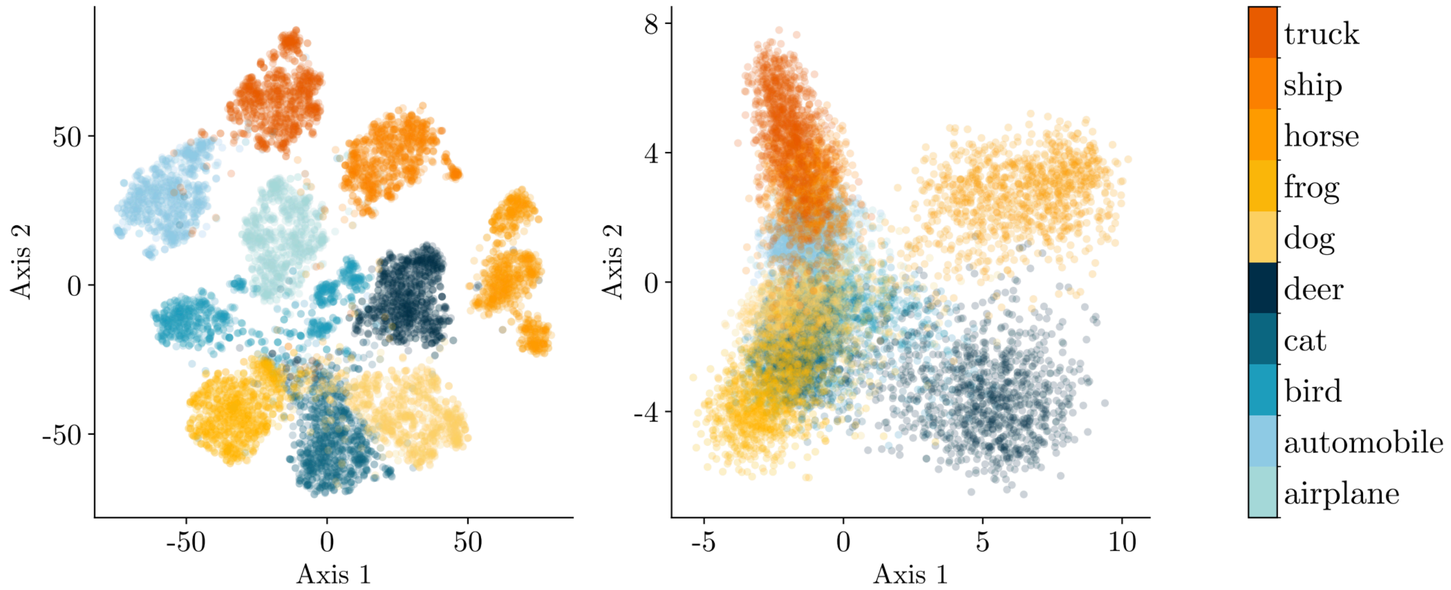

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Can we capture the minimal number of degrees of freedom this manifold has?

Can geometry provide label-free insights into representations' quality?

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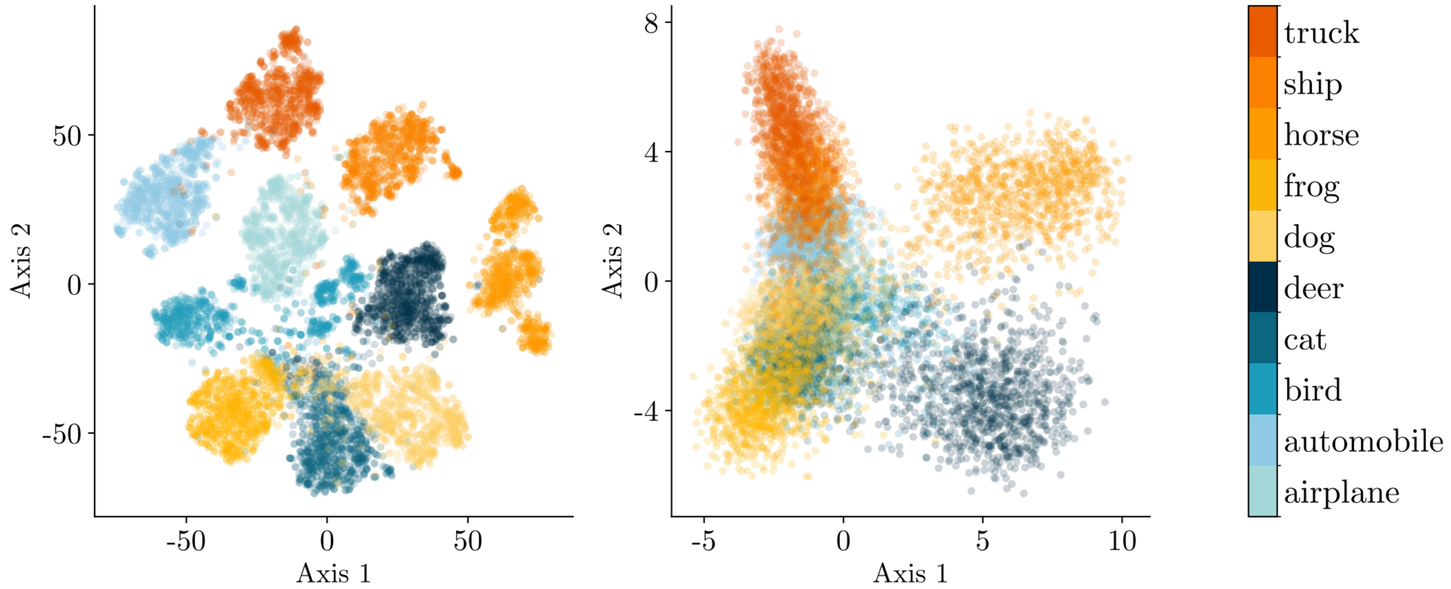


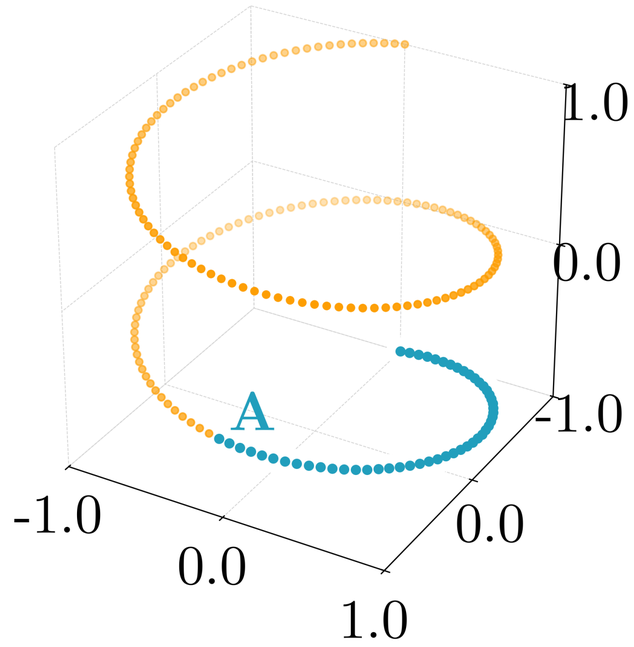
Dimension Reduction on CIFAR-10 with DINOv3

Can we capture the minimal number of degrees of freedom this manifold has?
Intrinsic Dimension Estimation

Can geometry provide label-free insights into representations' quality?

Intrinsic dimension:

« *characterises the effective number of degrees of freedom needed to represent data in an embedding », Bennett 1969*



The Intrinsic Dimensionality of signal collections, Transactions on Information Theory, Bennett, 1969

Why intrinsic dimension?

Existing Works on ID:
LLMs & Vision

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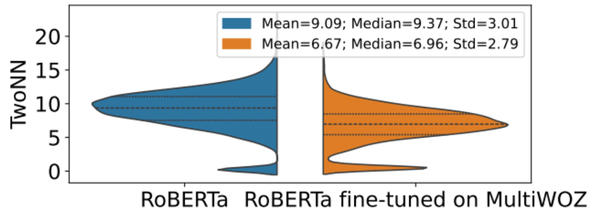
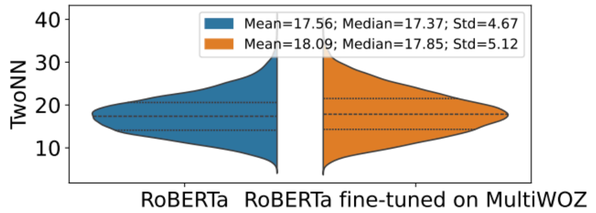


(a) TwoNN estimates on `MultiWOZ` validation



(b) TwoNN estimates on `Wikipedia` validation

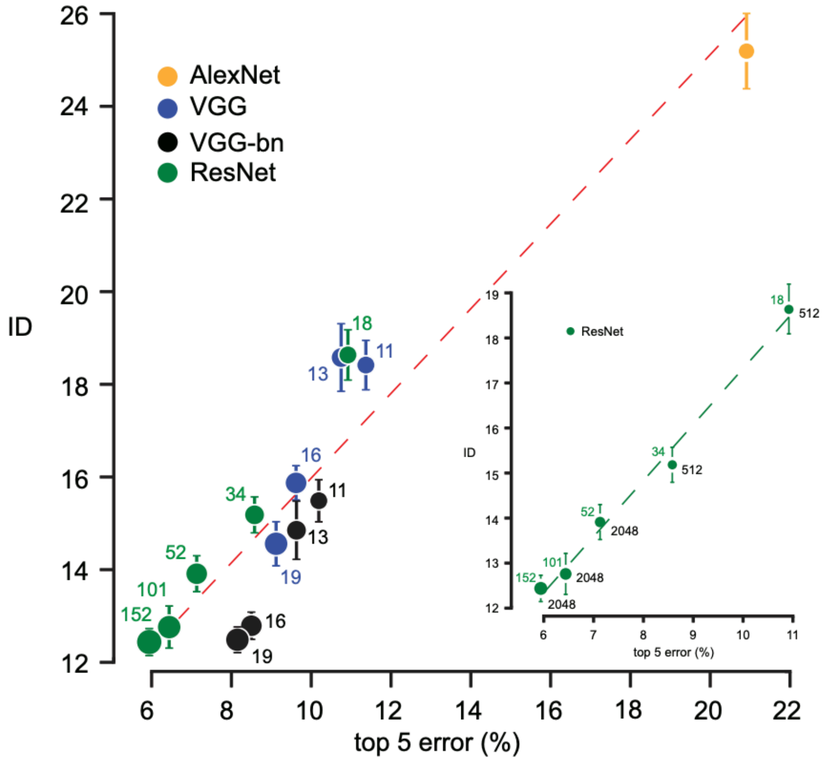
Fine-Tuning Induces Dataset-Specific Shifts
in Heterogeneous Local Dimensions

ID provides critical insights into generalisation

Less is More: Local Intrinsic Dimensions of Contextual Language Models, Ruppik et al, NeurIPS 2025

Why intrinsic dimension?

Existing Works on ID:
LLMs & Vision



Focused on Supervised CNNs

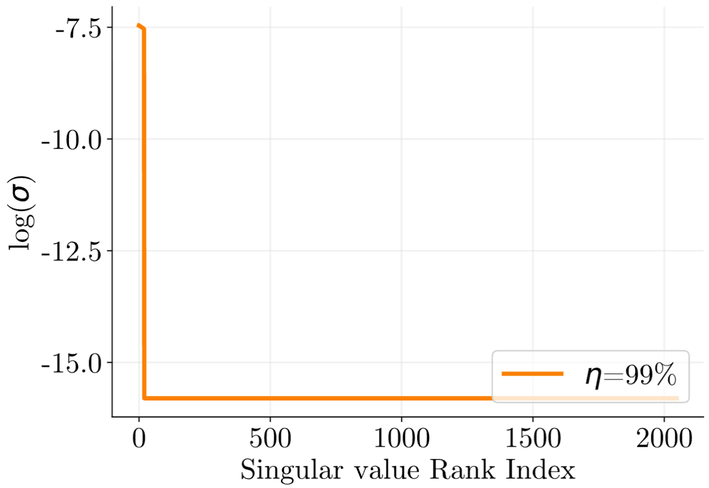
Intrinsic dimension of data representations in deep neural networks, Ansuini et al, NeurIPS 2019

Why intrinsic dimension?

Existing Works on ID:
LLMs & Vision

Go beyond existing methods tailored for collapse

Representation spectrum (RankMe, LiDAR, α-REQ)



Singular value spectrum
10,000 data points in dimension 2048 from an isotropic Gaussian distribution, then zero out a fraction η of their coordinates to control the collapse level

RankMe, Garrido et al, ICML 2023 · α-ReQ, Agrawal et al, NeurIPS 2022 · LiDAR, Thilak et al, ICLR 2024

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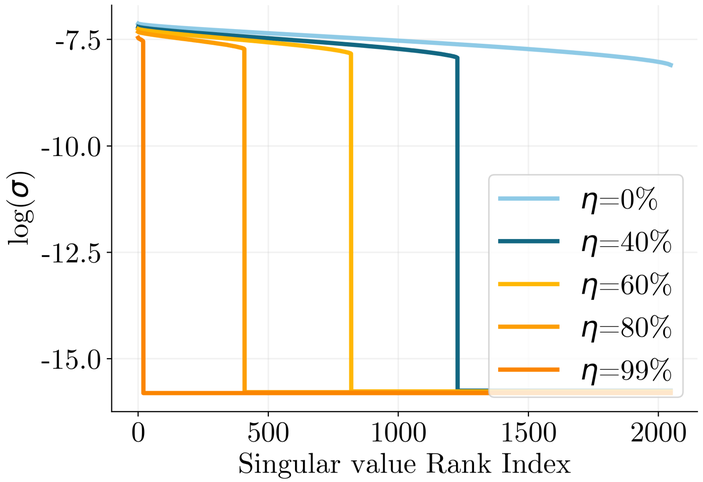

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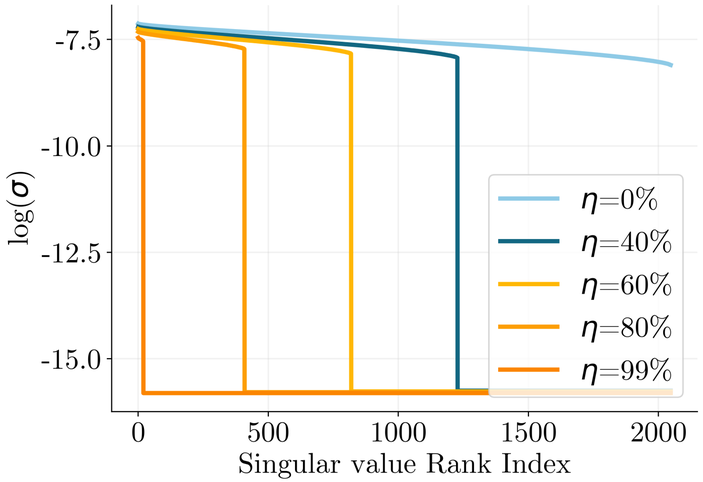
$$\text{RankMe}(Z) = \exp\Big(-\sum^{\min(N,K)} p_k \log p_k\Big) \text{ ,with } p_k = \frac{\sigma_k}{||\sigma_k||} + \epsilon$$



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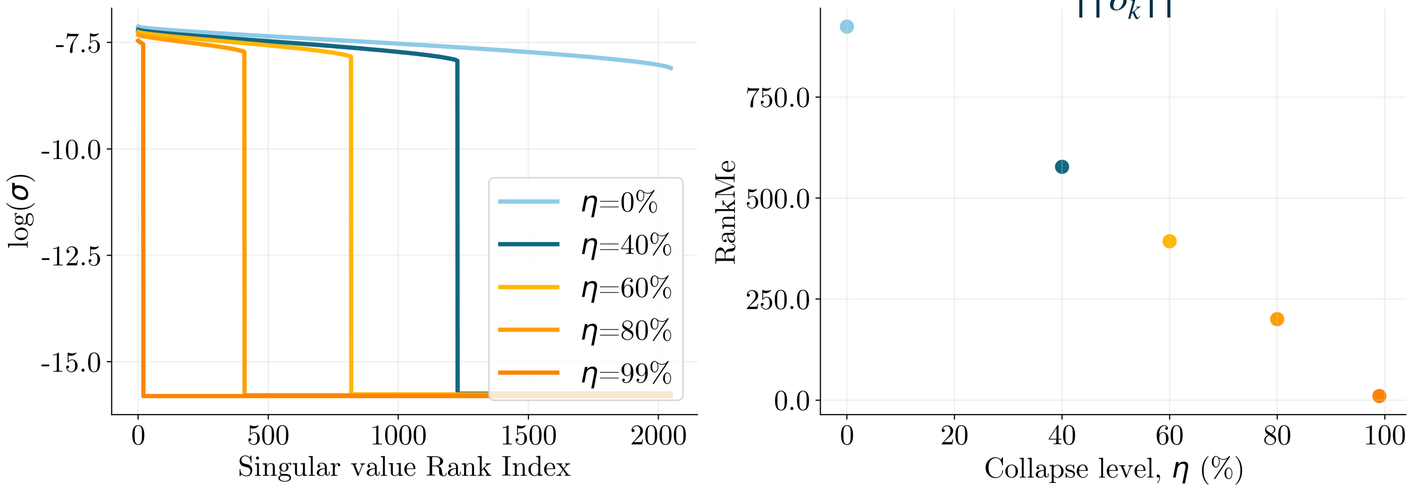


Singular value spectrum
10,000 data points in dimension 2048 from an isotropic Gaussian distribution, then zero out a fraction η of their coordinates to control the collapse level

Targets a specific failure mode of JE-SSL methods:
Are there other metrics that reflect performance beyond this apparent failure mode?

RankMe, Garrido et al, ICML 2023 · α-ReQ, Agrawal et al, NeurIPS 2022 · LiDAR, Thilak et al, ICLR 2024

Why intrinsic dimension?

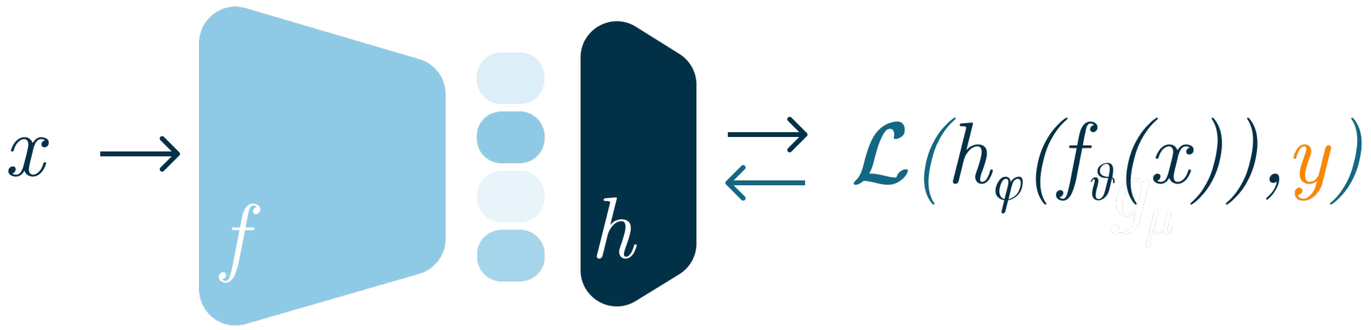
Existing Works on ID:
LLMs & Vision

Go beyond representation
spectrum tailored for collapse

Theoretical connection to generalisation

Let f a frozen **encoder**, mapping input to a d-dimensional latent manifold (e.g., SSL backbone), h a **classification head** mapping representations to classes and $\mathscr{L}$ a non-negative **loss function** (e.g., cross-entropy) satisfying $\mathscr{L}(f(x), \mathscr{F}(x)) = 0$ if $f(x) = \mathscr{F}(x)$

Theorem [Konz and Mazurowski, ICLR 2024]:
Let $K_{\mathscr{L}}$ the Lipschitz constant of the loss function. Then $\mathscr{L} \approx \mathcal{O}(K_L N_D^{-\frac{1}{d}})$



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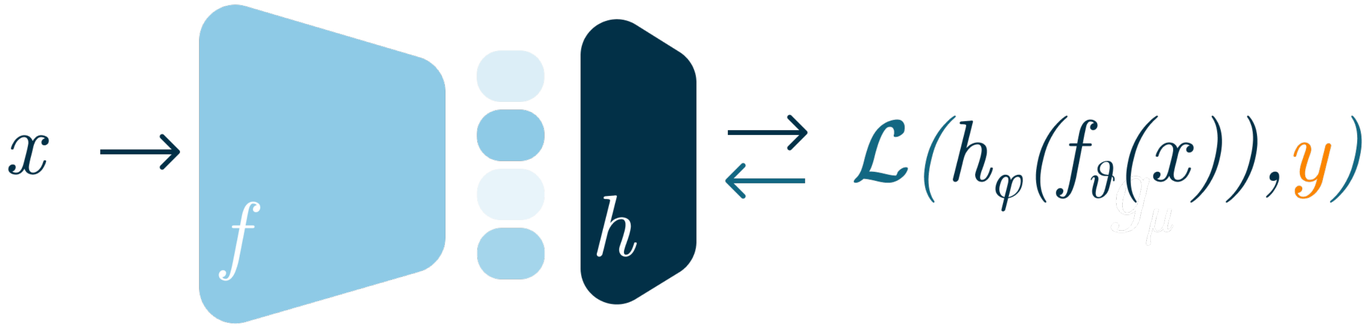

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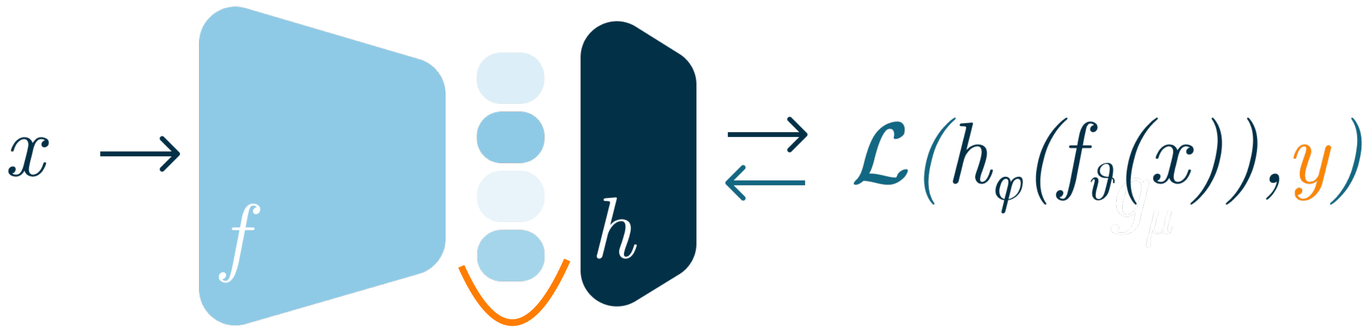
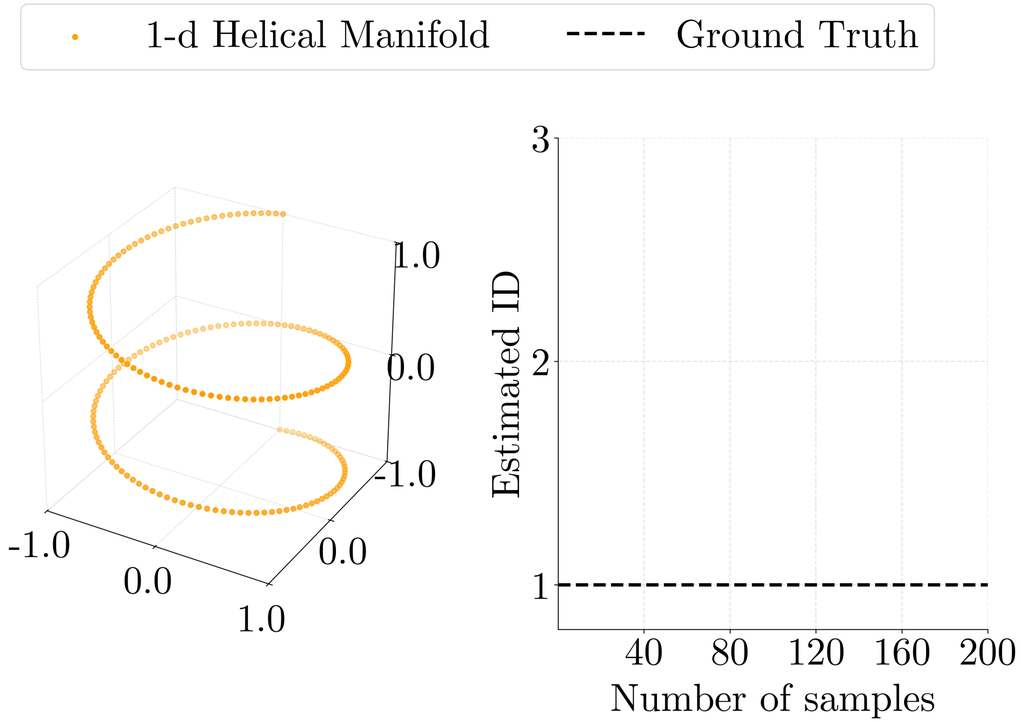


Capturing Intrinsic Dimension

Assuming the data points are sampled on (or close to) some unknown d-submanifold of the ambient space, **the goal is to estimate d from the data**



Capturing Intrinsic Dimension

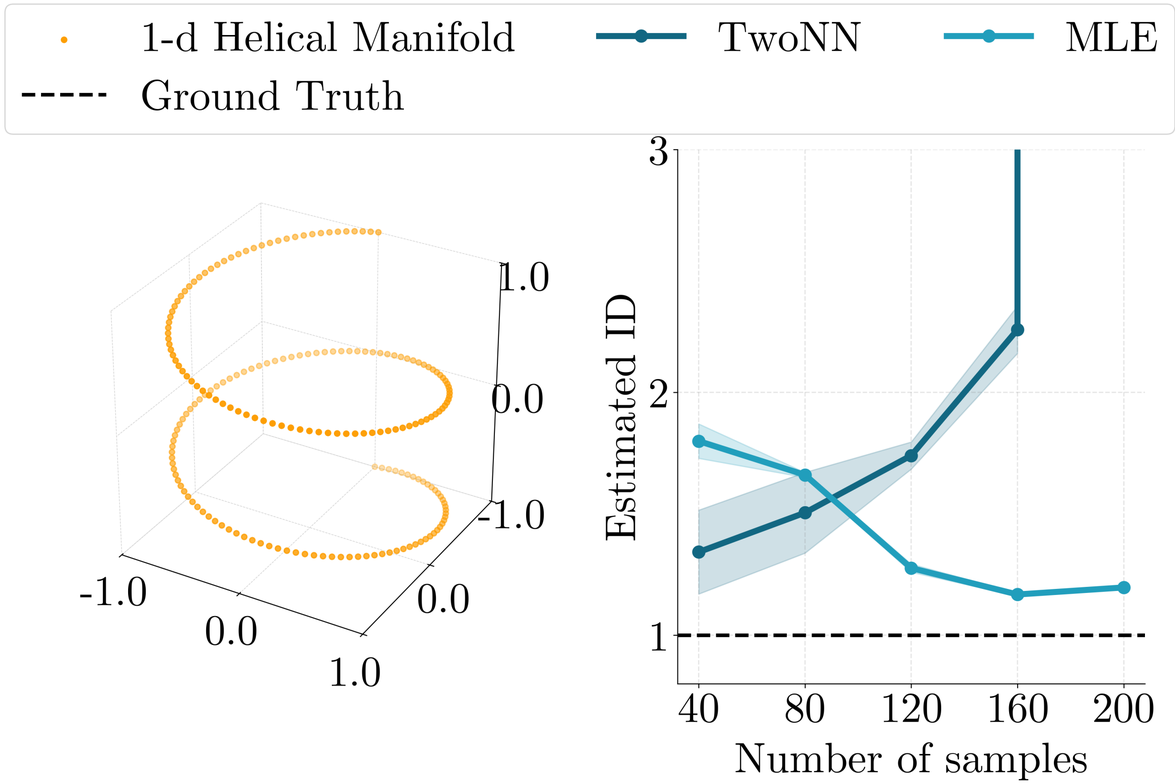
Parametric Estimators (MLE, TwoNN)

Assume i.i.d. sampling from distributions with locally constant density.
They model data as a local homogeneous Poisson process, where **neighbour distance ratios** depend on the intrinsic dimension d.

1. Unstable in heavy tail regimes
2. Sensitive to noise

Capturing Intrinsic Dimension

Parametric Estimators (MLE, TwoNN)



Capturing Intrinsic Dimension

Metric Invariant based estimators ($\dim_{\mathrm{MST}}$)

Relies on weak assumptions: compactness of the manifold and boundness of the Lebesgue sampling density supported on the manifold

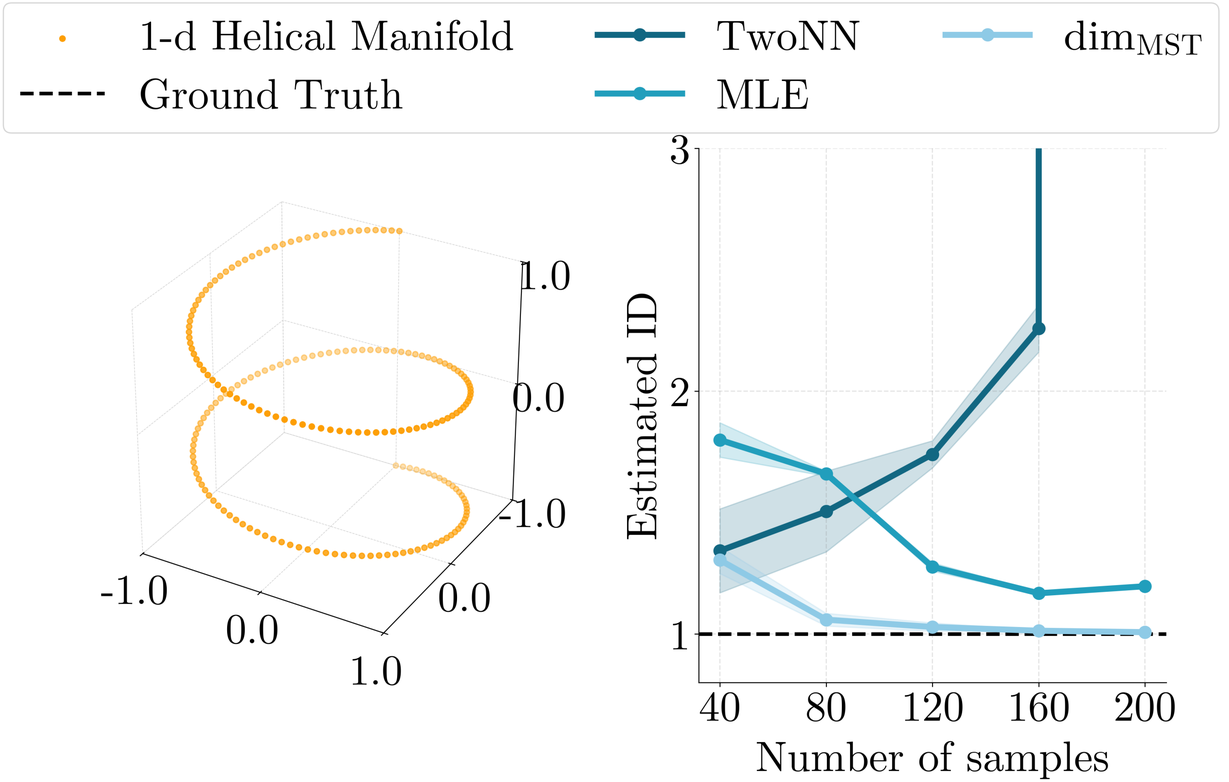
Algorithm 1 Computation of $\dim_{\mathrm{MST}}$

Input: The set of representations $Z = \{z_1, \ldots, z_N\}$, minimun sample size $n_{\min}$, skip size Δ
Output: The estimated dimension: $\dim_{\mathrm{MST}}(Z)$
Initialize $n \leftarrow n_{\min}$, $E \leftarrow []$
while $n < N$ **do**
  $Z_n \leftarrow \mathrm{sample}(Z, n)$ # random sampling of n points
  $E[i] \leftarrow L\left(\mathrm{MST}(Z_n)\right)$ # computation of the minimum spanning tree
  $n \leftarrow n + \Delta$
end while
$m, b \leftarrow$ linear regression$(\log(n_{\min} : \Delta : N), \log(E))$ # linear regression of the log-log plot
$\dim_{\mathrm{MST}}(\mathrm{Z}) \leftarrow 1/(1-m)$

Determining Intrinsic Dimension and Entropy of High-Dimensional Shape Spaces, Costa and Hero, Statistics and Analysis of Shapes, 2006

Capturing Intrinsic Dimension

Metric Invariant based estimators ($\dim_{\mathrm{MST}}$)



IdEst

Leverage $\dim_{\mathrm{MST}}$ to assess self-supervised representations:

1. *Can IdEst yield insights **across pretrained SSL models?***
2. *Can IdEst yield insights **along** self-supervised **pretraining**?*
3. *Can IdEst be leveraged for **unsupervised hyperparameter selection**?*

IdEst

1. Can IdEst yield insights across pretrained SSL models?

Intra-Dataset Correlation on ImageNet

Set-Up:

1. **Foundation Models: 35** models spanning **13 methods** across architecture (e.g., DINO ViT-S, DINOv2 ViT-S, CLIP RN-50)
2. **Compute IdEst and Linear Probe Accuracy on ImageNet**
3. **Correlation Coefficients between the given metric and linear probe accuracy**
Kendall's tau (τ) and Pearson (ρ)

IdEst

1. Can IdEst yield insights across pretrained SSL models?

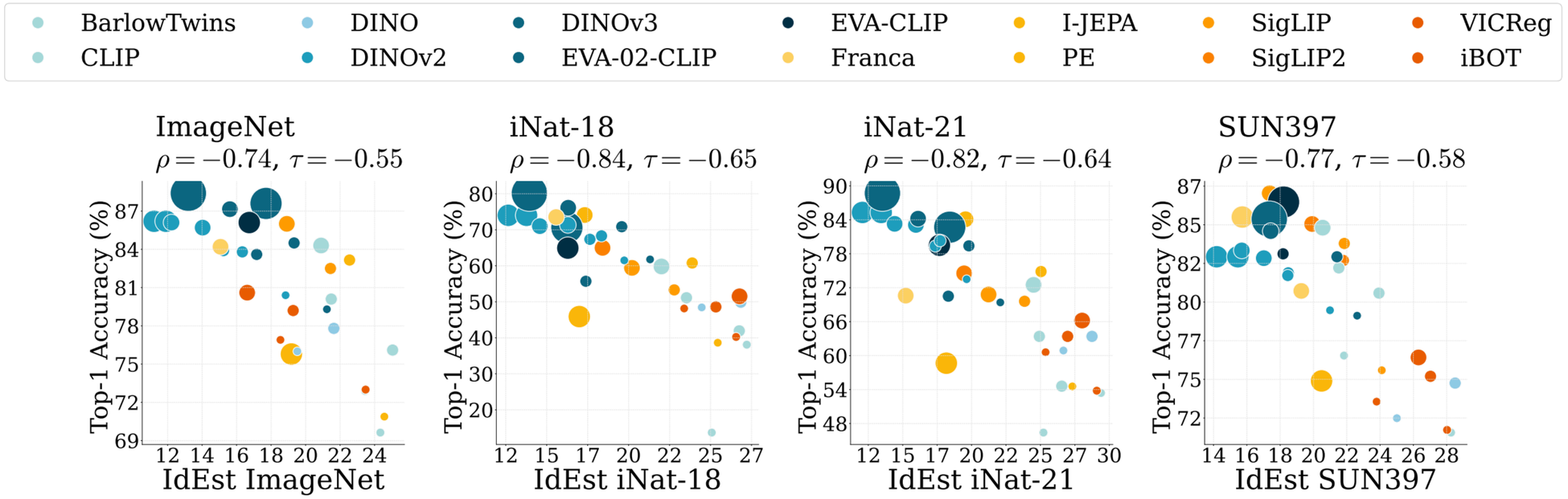
Intra-Dataset Correlation on ImageNet

Finding 1. IdEst negatively correlates with linear probing accuracy

IdEst

1. Can IdEst yield insights across pretrained SSL models?

Intra-Dataset Correlation beyond ImageNet

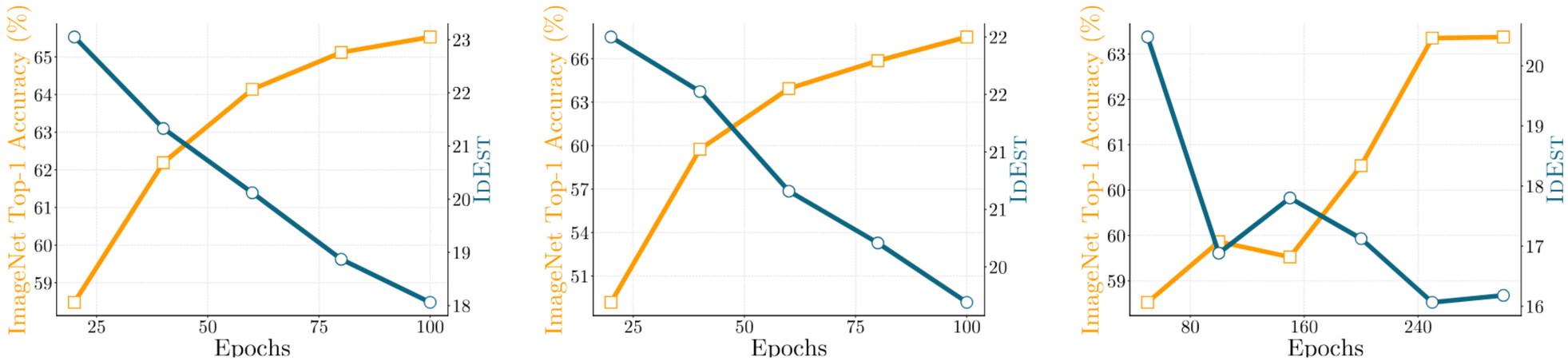


Finding 1. IdEst negatively correlates with linear probing accuracy

IdEst

2. Can IdEst yield insight along self-supervised pretraining?

Set-Up:
Evolution of ImageNet-1k linear probing top-1 accuracy (left y-axis) and **IdEst** (right y-axis) over self-supervised pretraining epochs

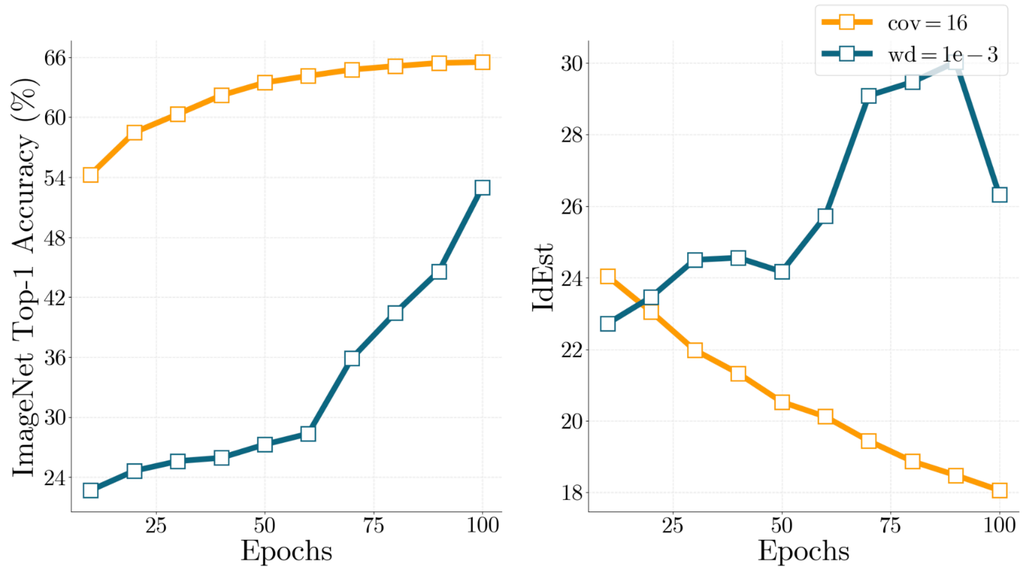


Finding 2. IdEst can serve as a label-free proxy for SSL pretraining effectiveness

IdEst

2. Can IdEst yield insight along self-supervised pretraining?

Set-Up:
Evolution of ImageNet-1k linear probing top-1 accuracy (left y-axis) and **IdEst** (right y-axis) over self-supervised pretraining epochs for one method and various hyparameters



Finding 2. IdEst can serve as a label-free proxy for SSL pretraining effectiveness

IdEst

3. Can IdEst be leveraged for unsupervised hyperparameter selection?

Set-Up:
Given a set of candidate models $\mathcal{F} = \{f_1, \ldots, f_n\}$ each trained with varying hyperparameters,
and their corresponding dimension estimates $\Delta = \{d_1, \ldots, d_n\}$,
the selected model, f, is defined as: $f^* = f_{\arg\min_i d_i}$

IdEst

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Given a method we train several models with varying hyperparameters configurations, e.g.,
I-JEPA (ViT-B)
- lr $\in \{4e-5, 8e-5, \ldots, 2e-4, 3e-4\}$
- target block size $\in \{\{0.075, 0.2\}, \ldots, \{0.2, 0.25\}\}$

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	I-JEPA (ViT-B)		
Method	lr	target-size	context-size
ACC-1 Bounds	[61.9, 66.4]	[49.0, 66.4]	[61.3, 66.5]
α-ReQ	61.9	49.0	61.3
RankMe	61.9	55.9	61.3
LiDAR	63.4	**66.4**	**66.0**
IdEst	**66.4**	**66.4**	**66.0**

Finding 3. IdEst can serve as an efficient label-free criterion for hyperparameter selection

IdEst

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	I-JEPA (ViT-B)			
Method	lr	target-size	context-size	all
ACC-1 Bounds	[61.9, 66.4]	[49.0, 66.4]	[61.3, 66.5]	[49.0, 66.5]
α-ReQ	61.9	49.0	61.3	49.0
RankMe	61.9	55.9	61.3	61.9
LiDAR	63.4	**66.4**	**66.0**	63.4
IdEst	**66.4**	**66.4**	**66.0**	**66.4**

Finding 3. IdEst can serve as an efficient label-free criterion for hyperparameter selection

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	VICReg(RN-50)	**DINO**(ViT-S)	**I-JEPA**(ViT-B)
Method	all	all	all
ACC-1 Bounds	[37.5, 69.1]	[48.4,69.1]	[49.0, 66.5]
α-ReQ	53.5	58.6	49.0
RankMe	**69.1**	63.6	61.9
LiDAR	65.0	**65.5**	63.4
IdEst	65.5	**65.5**	**66.4**

Finding 3. IdEst can serve as an efficient label-free criterion for hyperparameter selection

IdEst

Computational cost

Model	Architecture	D	Param (M)	Linear Probe (min)	IdEst (min)
VICReg	ResNet-50	2048	24	64.5	2.1
DINOv2	ViT-S	384	22	65.8	1.8
	ViT-B	768	86	64.5	2.1
	ViT-L	1024	303	113.3	4.9
	ViT-G	1408	1000	322.7	12.8

IdEst reduces computational overhead,
achieving up to **30x** speedup over linear probing

Conclusion & Perspectives

Conclusion

Capturing geometry, a powerful lens to study SSL representations:

1. Shaping Representations

Preventing known failure modes during training

T-REGS:
Enforcing uniformity to bypass collapse in joint-embedding self-supervised learning

2. Evaluating Representations

Studying quality beyond downstream performances

IdEst:
Studying representations via intrinsic dimension

On Minimum Spanning Trees

```python
from scipy.spatial.distance import pdist, squareform
from scipy.sparse.csgraph import minimum_spanning_tree

dists = squareform(pdist(z))  # pairwise distances
mst = minimum_spanning_tree(dists)
length =  mst.sum()
```

MST length via **scipy**

```python
from ripser import ripser

pd = ripser(z, maxdim=0)["dgms"][0]
# remove the infinite bar (last connected component)
finite = pd[pd[:, 1] != np.inf]
length = (finite[:, 1] - finite[:, 0]).sum()
```

MST length via **ripser**

Future Directions

Leveraging T-REGS for Tammes Problem

Dimension	Number of Points	Cosinus of minimal angle	Reference
3	7*	0.210138312731	A Schütte et al, uf welcher Kugel haben 5, 6, 7, 8 oder 9 Punkte mit Mindestabstand Eins Platz?, 1951
3	8*	0.261203874964	A Schütte et al, uf welcher Kugel haben 5, 6, 7, 8 oder 9 Punkte mit Mindestabstand Eins Platz?, 1951

* optimal solutions

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		...	
3	131	0.948765511183	Lai et al, Iterated dynamic neighborhood search for packing equal circles on a sphere, Comput. Oper. Res., 2023
3	132	0.948962593971	Lai et al, Iterated dynamic neighborhood search for packing equal circles on a sphere, Comput. Oper. Res., 2023
3	133	0.949572658198	Lai et al, Iterated dynamic neighborhood search for packing equal circles on a sphere, Comput. Oper. Res., 2023
3	134	0.949971678353	Lai et al, Iterated dynamic neighborhood search for packing equal circles on a sphere, Comput. Oper. Res., 2023
3	135	0.950317557456	Lai et al, Iterated dynamic neighborhood search for packing equal circles on a sphere, Comput. Oper. Res., 2023

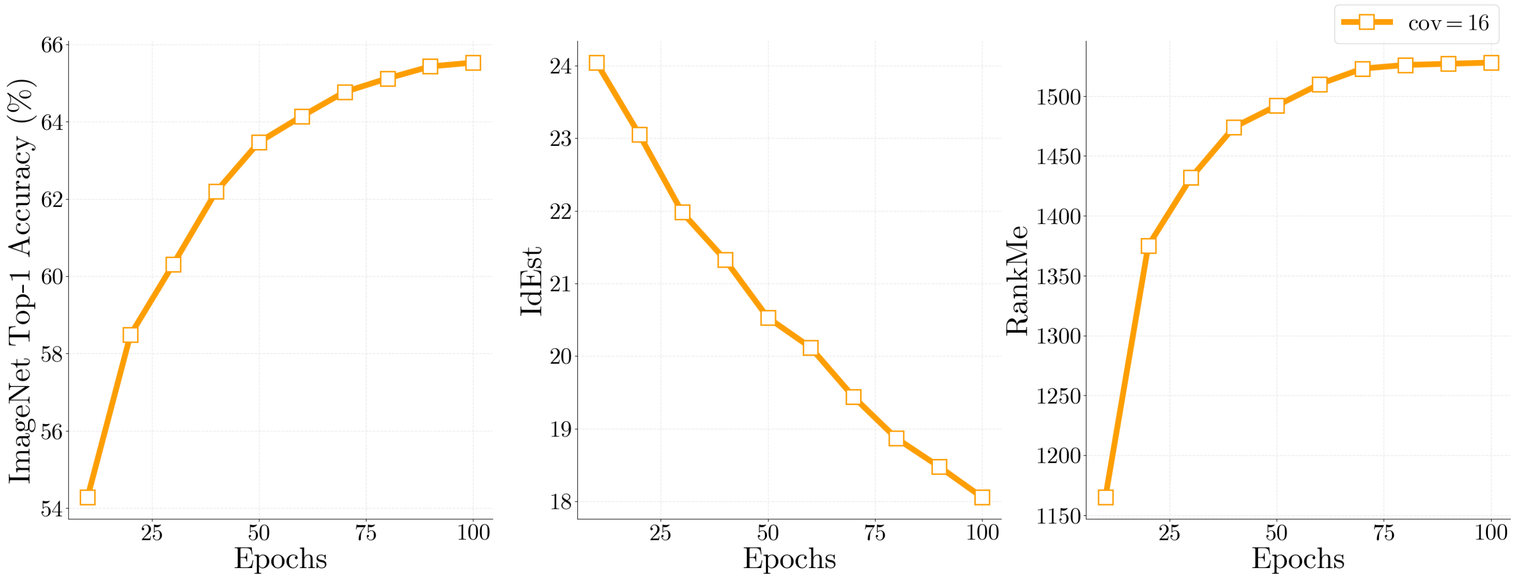
* optimal solutions

Henry Cohn, Spherical codes

Future Directions

Leveraging T-REGS for Tammes Problem

On high rank and low intrinsic dimension for deeper understanding



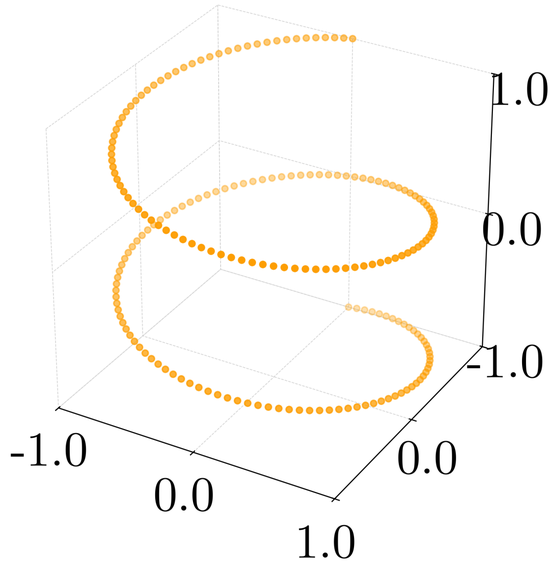
ImageNet Acc, IdEst, RankMe along VICReg Pretrainig

Future Directions

Leveraging T-REGS for Tammes Problem

On high rank and low intrinsic dimension for deeper understanding

Information Bottleneck:
model squeezes out redundancy (low ID) while retaining as much information as possible (high rank)



A curved space maintains both low ID and high rank: differential geometry

Beyond vision SSL

Can capturing geometry be leveraged further?

Generative Modelling

Exploring how geometric priors and structural representations can enhance generative architectures and visual synthesis.

Other Modalities

Extending geometrically-grounded self-supervised learning principles to 3D, audio, and multi-modal domains.

Thank you!

Appendices

T-REGS

T-REGS

MST & Uniformity: sparse regime ($n \approx d$)

$n \leq d + 1$, e.g. batch sizes are often smaller than or comparable to the ambient dimension ; assume Z lies in a closed Euclidean d-ball B of radius r centered at the origin.

Theorem [M., Loiseaux, Kalogeiton, Oudot, 2025]: Under the above conditions, the maximum of the $E(\mathrm{MST}(Z))$, over the point sets $X \subset B$ of fixed cardinality n is attained when the points of X lie on the sphere $S = \partial B$, at the vertices of a regular $(n-1)$-simplex that has S as its smallest circumscribing sphere.

The proof relies on two main ingredients:

i) **Standard results in convex geometry:**
Under the conditions of the above theorem,
$\sum_{1 \leq i < j \leq n} \| z_i - z_j \|_2$ is maximal when the points of X lie on the bounding sphere S, at the vertices of a regular d-simplex

ii) **Technical lemma: relating the length of the MST to sum of pairwise distances:**
For any points $z_1, \ldots, z_n \in \mathbb{R}^d$: $E\left(\mathrm{MST}\left(\{z_1, \ldots, z_n\}\right)\right) \leq \frac{2}{n} \sum_{1 \leq i < j \leq n} \| z_i - z_j \|_2$

T. M. Apostol and M. A. Mnatsakanian. New horizons in geometry, volume 47, 2017

T-REGS

Soft vs hard sphere constraint

Coefficients			Scaling		
β	γ	λ	$\frac{\beta}{\gamma}$	$\frac{\gamma}{\lambda}$	**Top-1**
1	-	-	-	-	collapse
1	1	-	1	-	collapse
10	1	1	10	1	collapse
10	0.5	5e-2	20	10	25.7
10	0.2	2e-2	50	10	45.4
10	0.5	2.5e-3	20	200	65.0
10	0.2	1e-3	50	200	65.3
10	0.5	2e-3	20	250	64.9
10	0.2	8e-4	50	250	**66.1**
10	0.02	8e-5	100	300	63.3

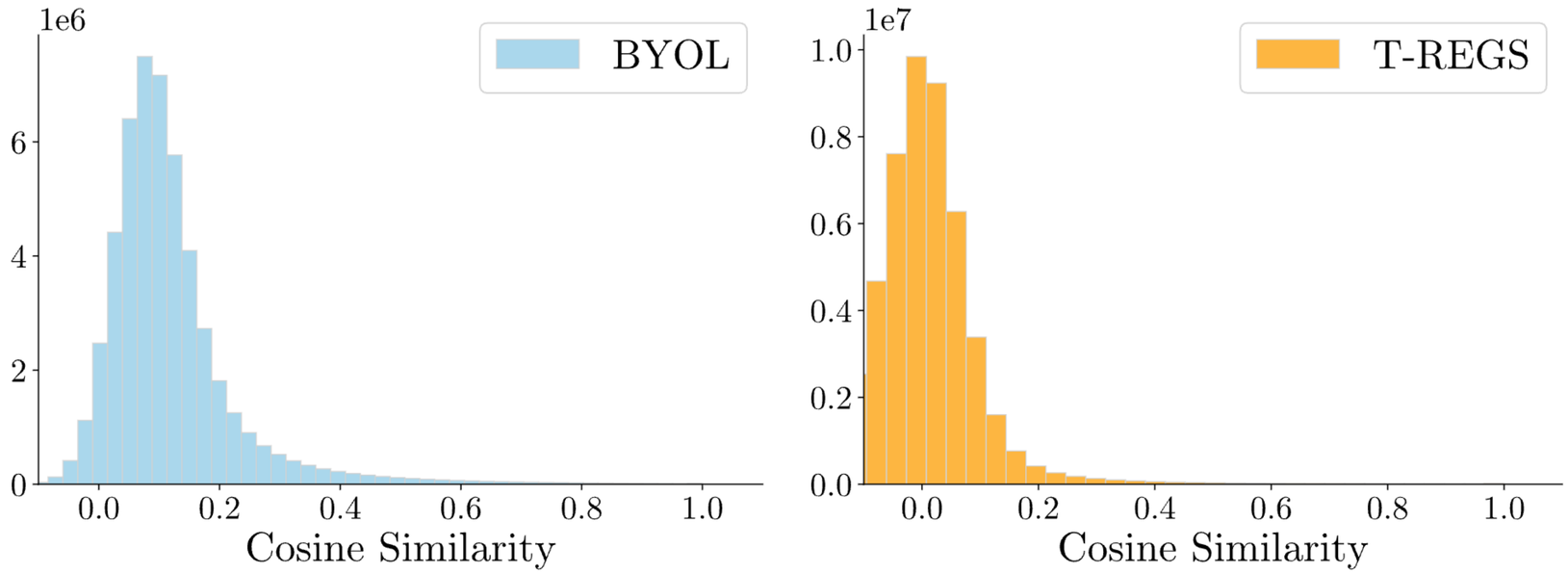
Table 4: **Impact of coefficients.** $\mathcal{L}_{\text{MSE}}$ $+\mathcal{L}_{\text{T-REGS}}$ top-1 accuracy (%) on ImageNet-1k with online evaluation protocol over 50 epochs. **Boldface** indicates best performance.

	CIFAR-10	**CIFAR-100**
soft-constraint	91.2	66.8
hard-constraint	89.2	64.7

Table 8: **Impact of normalization.** $\mathcal{L}_{\text{MSE}}$ $+\mathcal{L}_{\text{T-REGS}}$ top-1 accuracy (%).

T-REGS

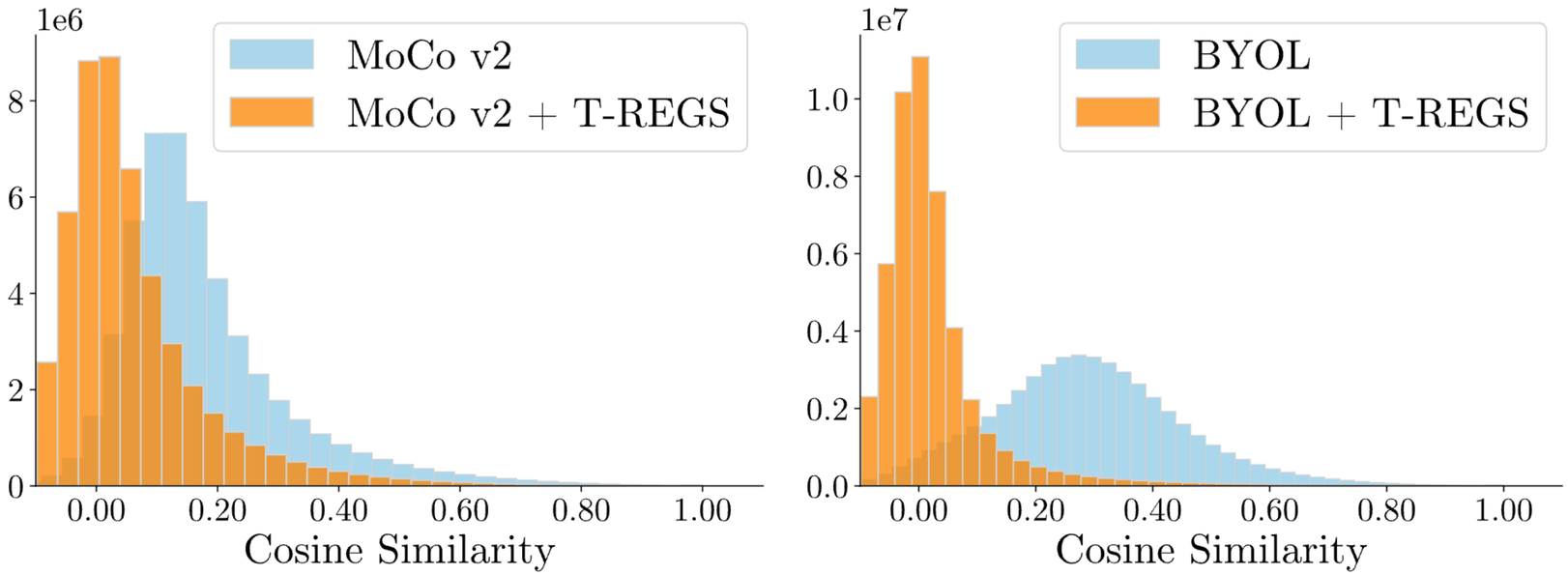
Evaluation beyond linear probing: geometric properties



Histogram of cosine similarities on **ImageNet-1k**

T-REGS

Evaluation beyond linear probing: geometric properties

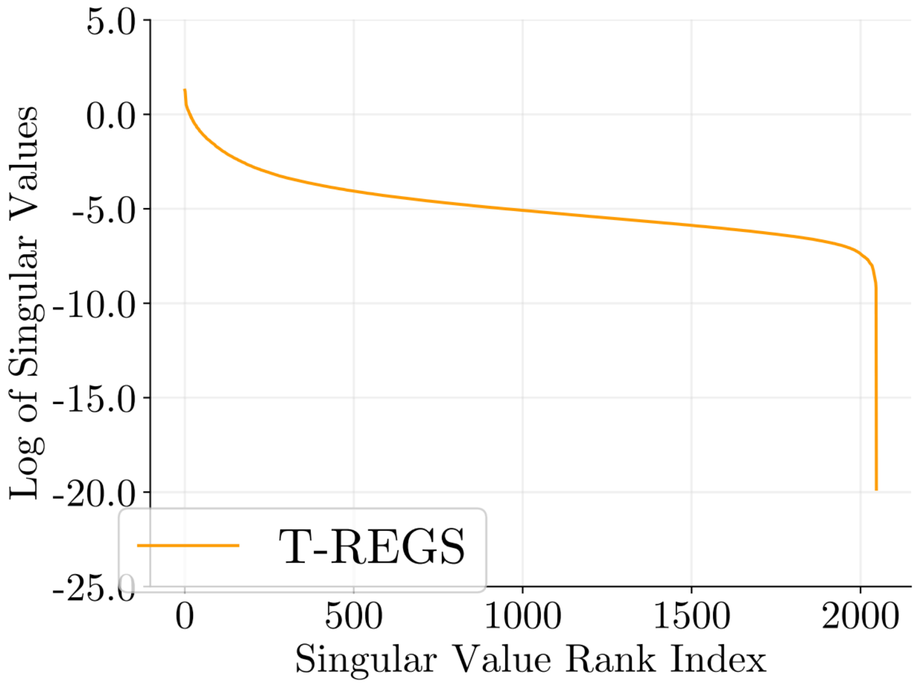


Histogram of cosine similarities on **CIFAR-10**

T-REGS

Evaluation beyond linear probing: geometric properties

Singular value spectrum of the representations

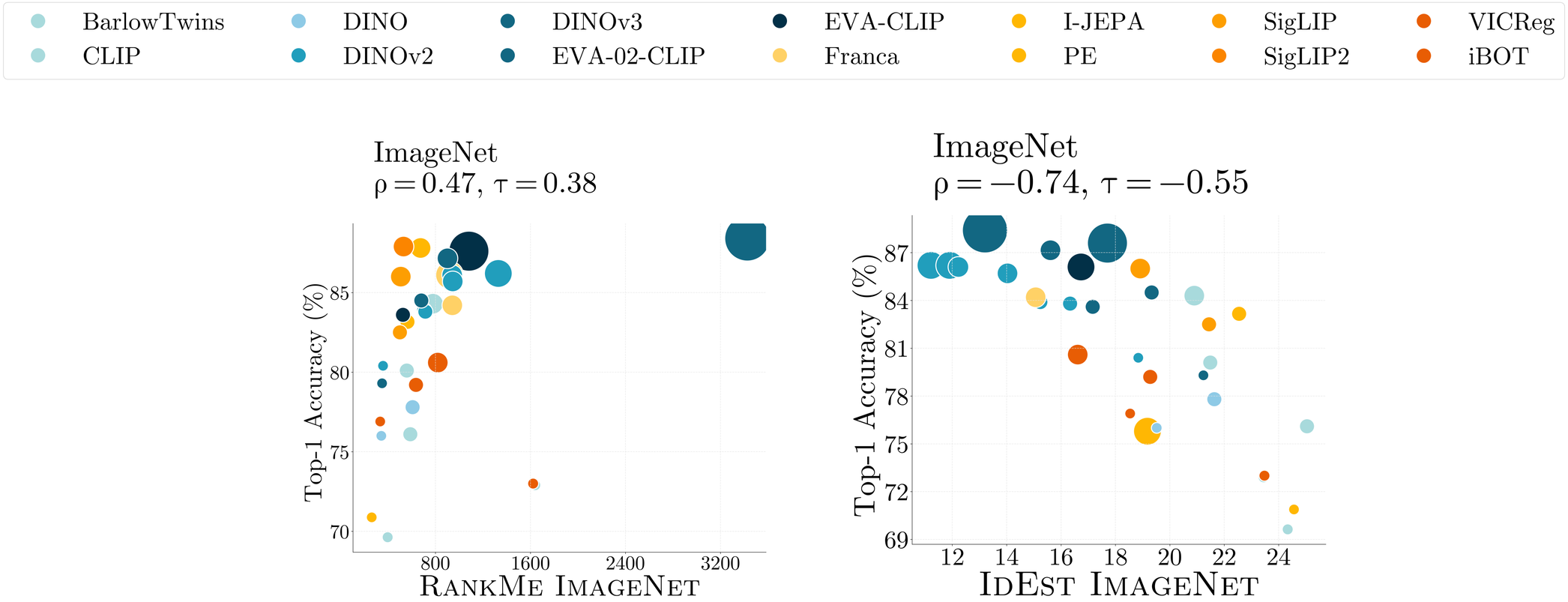


Singular value spectrum of the representation space on ImageNet-1k

IdEst

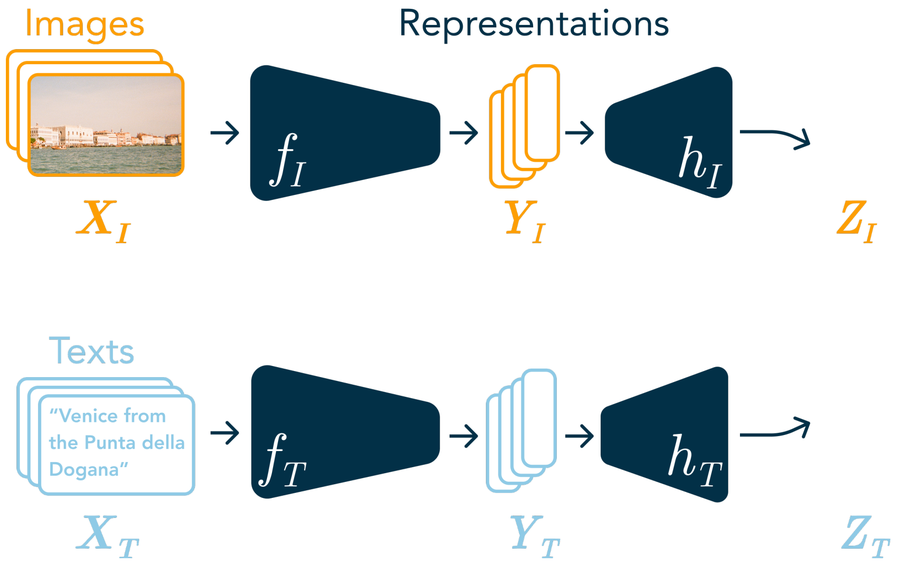
IdEst

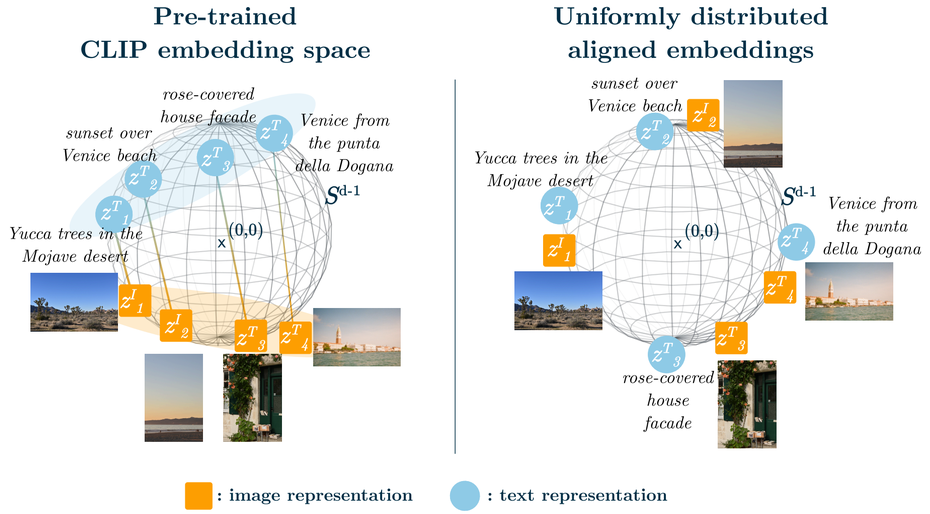
On high rank and low intrinsic dimension



ADAPT

Multimodal Joint-Embedding Architectures





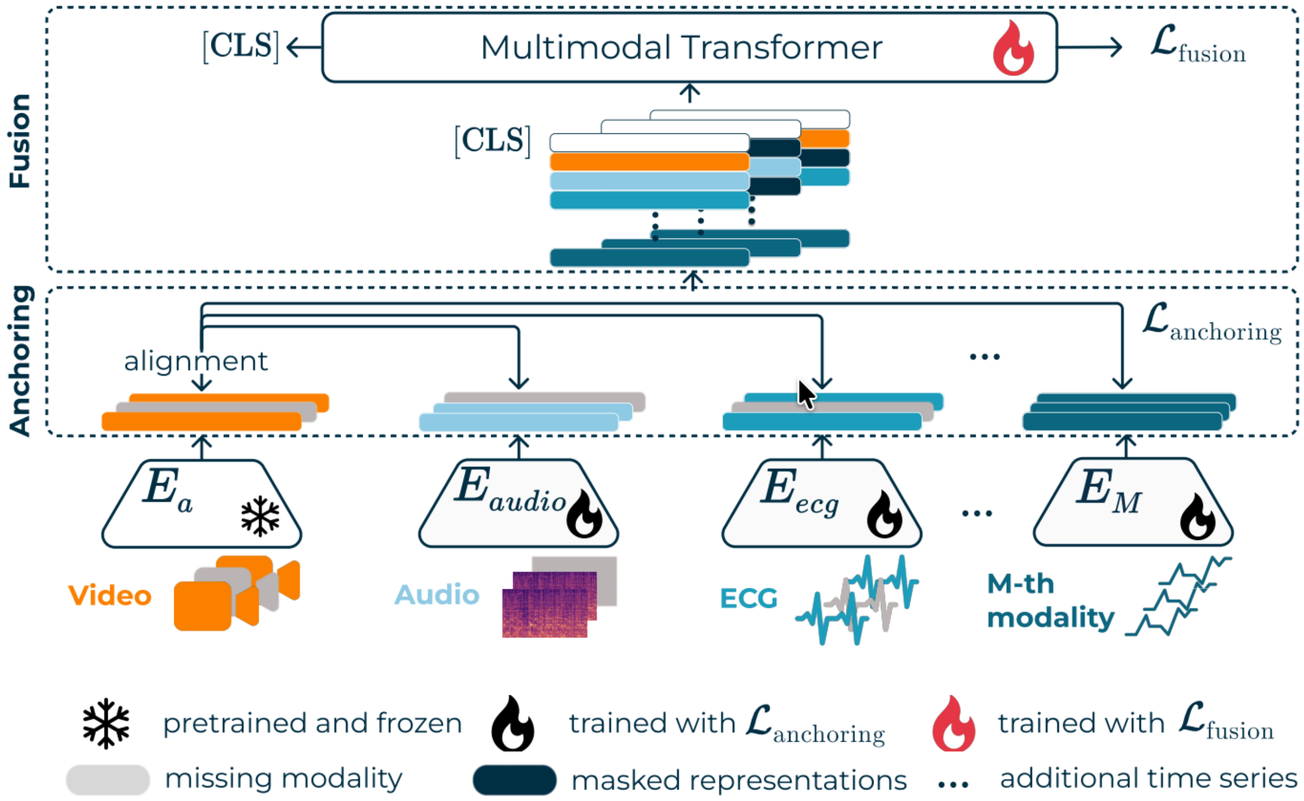
Method	Flickr30k [49] i → t R@1	Flickr30k [49] i → t R@5	Flickr30k [49] t → i R@1	Flickr30k [49] t → i R@5	MS-COCO [43] t → i R@1	MS-COCO [43] t → i R@5	MS-COCO [43] t → i R@1	MS-COCO [43] t → i R@5
Zero-Shot	71.1	90.4	68.5	88.9	31.9	56.9	28.5	53.1
Finetune	81.2	95.5	80.7	95.8	36.7	63.6	36.9	63.9
ES [41]	71.8	90.0	68.5	88.9	31.9	56.9	28.7	53.0
i-Mix [39]	72.3	91.7	69.0	91.1	34.0	63.0	34.6	62.2
Un-Mix [54]	78.5	95.4	74.1	91.8	38.8	66.2	33.4	61.0
m^3-Mix [45]	82.3	95.9	**82.7**	96.0	41.0	68.3	39.9	67.9
$\mathcal{L}_{\text{CLIP}} + \mathcal{L}_{\text{T-REGS}}$	**83.2**	**96.0**	80.8	**96.4**	**41.6**	**68.7**	**41.5**	**68.7**

Geodesic Multi-Modal Mixup for Robust Fine-Tuning, Oh et al, NeurIPS 2023

CLIP, Radford et al, ICML 2021 ·

ADAPT

Fusing modalities: beyond bi-modal datasets with T-REGS



	ACC	F1
Unimodal		
Video	62(4)*	67(3)*
Biomedical signals	58(4)*	66(5)*
Audio	62(4)*	67(4)*
Multimodal		
Feature Fusion [28]	61(3)*	66(4)*
Decision-level fusion [28]	65(5)*	72(5)*
ShaSpec+ [195]	70.2(3.7)	75.7(5.3)
ADAPT	**71.3(4.2)**	**78.7(4.6)**

	X^*_{test}		X_{test}	
T-REGS	**ACC**	**F1**	**ACC**	**F1**
×	69.9(3.7)	77.0(4.3)	72.9(2.9)	72.9(3.1)
✓	**71.3(4.2)**	**78.7(4.6)**	**74.6(1.9)**	**74.7(2.0)**

Table 4.4: **Ablation of the T-REGS term in Eq. (4.4) on X^*_{test} and X_{test}.** We compare training with $\mathcal{L}_{\text{anchoring}}$ alone (×) versus the full objective that adds $\sum_m \mathcal{L}_{\text{T-REGS}}(Z_m)$ (✓). **Bold** indicates the best result per column.